Dense Pixel-Wise Alignment of Infrared and Visible Imagery: A Mask-Guided Deep Learning Approach

A Research Survey and Proposed Methodology

Graduate Coursework

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This document surveys methods for dense, per-pixel registration of infrared (IR) and visible (RGB) imagery using deep learning. It presents a proposed architecture (IRAlign-Net) that predicts displacement fields with confidence masks, along with comprehensive coverage of augmentation strategies, loss functions, and training methodologies for the limited-data regime of a few thousand aligned video frames.

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Chapter 1

Dense Pixel-Wise Alignment Methods for Cross-Modal IR–RGB Registration

1.1 Introduction

Fusing infrared (IR) and visible (RGB) imagery is a prerequisite for numerous applications in surveillance, autonomous driving, and remote sensing. A critical first step is *registration*: bringing corresponding pixels of the two modalities into spatial agreement. Classical approaches estimate a global geometric transformation—an affine or projective homography—that maps one image onto the other. While adequate when the scene is planar or the camera baseline is negligible, a single global transformation cannot account for parallax, lens distortion differences between the two sensors, or local deformations introduced by independent optical paths.

This chapter surveys methods that instead predict a *dense displacement field* $\boldsymbol{\phi} : \mathbb{R}^2 \to \mathbb{R}^2$, assigning every pixel (u, v) an offset $(\Delta x, \Delta y)$ such that the warped pixel location is $(u + \Delta x,\ v + \Delta y)$. The resulting deformation is non-parametric—it can model arbitrary spatially varying misalignment. We organise the discussion around five families of architectures: optical-flow networks (Section 1.2), Spatial Transformer Networks (Section 1.3), deformable convolutions (Section 1.4), U-Net–based displacement regressors (Section 1.5), and the VoxelMorph registration framework (Section 1.6). For each, we describe the architecture, the mechanism by which per-pixel displacements are produced, and the specific advantages and limitations when the two images to be aligned exhibit the drastic appearance differences inherent to the IR–RGB pair.

1.2 Optical Flow–Based Approaches

Optical flow estimation seeks to recover the apparent 2D motion field between two temporally adjacent frames. Because the output is a dense (dx, dy) map at every pixel, the same machinery can in principle be repurposed for spatial registration of an IR–RGB pair.

1.2.1 FlowNet and FlowNet 2.0

FlowNet Dosovitskiy et al. [2015] was the first work to cast optical flow as a supervised learning problem solved by a convolutional neural network (CNN). Two variants were proposed: *FlowNetSimple* (FlowNetS), which concatenates the two input images along the channel dimension and feeds them through a contracting encoder followed by an expanding decoder with "upconvolutional" layers; and *FlowNetCorr* (FlowNetC), which processes each image through a separate feature tower and explicitly computes a correlation volume between the two feature maps before decoding. Both produce a dense flow field at the original resolution via successive upsampling and refinement steps. Training was performed on the synthetic *Flying Chairs* dataset.

FlowNet 2.0 Ilg et al. [2017] extended the original work with three key contributions: (i) a carefully designed training schedule that progressively increases data complexity, (ii) a stacked cascade of FlowNet modules in which each stage warps the second image by the current flow estimate before refining the residual, and (iii) a specialised sub-network for small displacements. The result was a halving of endpoint error relative to FlowNet while retaining real-time throughput.

1.2.2 PWC-Net

PWC-Net Sun et al. [2018] embeds three classical optical-flow principles—**P**yramid processing, **W**arping, and **C**ost-volume construction—into a compact, end-to-end trainable CNN. A learnable feature pyramid is constructed for both images. At each pyramid level the second image's features are warped by the current flow estimate, a partial cost volume is built between the warped features and the first image's features (limiting the displacement search range for efficiency), and a lightweight CNN refines the flow. The coarse-to-fine strategy allows the network to capture large displacements at coarse levels and fine detail at fine levels. PWC-Net is roughly 17× smaller than FlowNet 2.0 and achieved state-of-the-art accuracy on MPI Sintel and KITTI 2015 at the time of publication.

1.2.3 RAFT

RAFT (Recurrent All-Pairs Field Transforms) Teed and Deng [2020] marked a paradigm shift by replacing coarse-to-fine pyramids with an iterative, recurrent refinement operating on a single high-resolution correlation volume. The architecture comprises three components:

1. **Feature encoder.** A shared-weight residual network maps each input image to a feature map at 1/8 resolution ($H/8 \times W/8 \times 256$). A separate *context encoder* extracts features from the first image only.

2. **All-pairs correlation volume.** The dot product of every feature vector in image 1 with every feature vector in image 2 produces a 4D $H \times W \times H \times W$ volume. The last two dimensions are average-pooled at multiple scales, yielding a multi-scale correlation pyramid.

3. **Recurrent update operator.** A convolutional GRU iteratively refines the flow estimate. At each iteration it performs a lookup into the correlation pyramid around the current correspondence and updates the flow via a learned residual. Because

weights are shared across iterations, the number of refinement steps can be varied at inference time.

RAFT won the ECCV 2020 Best Paper Award and achieved a 16% error reduction on KITTI and 30% on Sintel (final pass) relative to prior published results, while using only 4.8 M parameters.

1.2.4 Adaptation to Cross-Modal IR–RGB Registration

Optical-flow networks are trained on pairs of images with consistent photometric appearance; the brightness constancy assumption is deeply embedded in the synthetic training data (Flying Chairs, FlyingThings3D, Sintel) and in photometric losses. When the two inputs are an IR and an RGB frame, this assumption is violated: a hot object may be bright in IR yet dark in RGB, and vice versa. Several strategies have been explored to bridge this domain gap:

- **Cross-modal image translation.** An auxiliary network translates the RGB image into a pseudo-IR image (or vice versa), reducing the problem to mono-modal flow estimation. Recent work by Bird et al. [2024] generates synthetic IR optical-flow datasets by applying an RGB-to-IR style transfer to existing flow benchmarks, enabling direct supervision.
- **Feature-level matching.** Rather than computing correlation on raw pixels, one can extract modality-invariant features (e.g. via a domain-adversarial encoder) and compute the cost volume in that shared feature space.
- **Structural similarity losses.** Replacing the photometric L_1 or L_2 reconstruction loss with modality-robust objectives such as mutual information, the structural similarity index (SSIM) computed on gradient magnitudes, or normalised cross-correlation can guide unsupervised fine-tuning on real IR–RGB pairs.

Limitations. Even with these adaptations, purely flow-based methods face two challenges: (i) they are designed for temporally adjacent frames and may struggle with the large appearance gap when no temporal cue exists, and (ii) publicly available pre-trained weights encode strong inductive biases toward same-modality matching, making domain transfer non-trivial.

1.3 Spatial Transformer Networks

1.3.1 The Original STN

The Spatial Transformer Network (STN), introduced by Jaderberg et al. Jaderberg et al. [2015b], provides a differentiable module that can be inserted into any CNN to perform learned spatial transformations of feature maps. The module consists of three components:

1. **Localisation network.** A small CNN (or fully connected network) that takes the input feature map and regresses the parameters $\boldsymbol{\theta}$ of the desired transformation (e.g. a 2×3 affine matrix).

2. **Grid generator.** Given $\boldsymbol{\theta}$, a regular grid of output coordinates is mapped back to input coordinates via $\begin{pmatrix} x_i^s \\ y_i^s \end{pmatrix} = \mathbf{A}_{\boldsymbol{\theta}} \begin{pmatrix} x_i^t \\ y_i^t \\ 1 \end{pmatrix}$, where (x_i^t, y_i^t) are target (output) grid coordinates and (x_i^s, y_i^s) are the corresponding source (input) sampling locations.

3. **Differentiable sampler.** Bilinear (or other sub-pixel) interpolation is applied at each (x_i^s, y_i^s) to produce the output pixel value. Crucially, the gradient of the interpolation kernel with respect to both the input feature map and the sampling coordinates is well-defined, enabling end-to-end back-propagation through the entire module.

The original paper demonstrated affine and projective transformations, but the framework is not limited to low-dimensional parametric families.

1.3.2 Extension to Dense Deformation Fields

The key insight enabling non-parametric warping is that the grid generator need not be parameterised by a compact $\boldsymbol{\theta}$. Instead, the localisation network can directly output a per-pixel displacement map $\boldsymbol{\phi} \in \mathbb{R}^{H \times W \times 2}$. The sampling grid then becomes

$$x_i^s = x_i^t + \phi_x(x_i^t, y_i^t), \qquad y_i^s = y_i^t + \phi_y(x_i^t, y_i^t),$$

where ϕ_x, ϕ_y are the predicted horizontal and vertical displacements. The differentiable bilinear sampler remains unchanged; gradients flow back through the displacements into the localisation network.

This dense variant underpins most modern learned registration methods. de Vos et al. de Vos et al. [2017] proposed **DIRNet**, which couples a convolutional regressor with a B-spline–parameterised spatial transformer, outputting a control-point grid that is upsampled to a full displacement field. Training is unsupervised, using an image-similarity loss (e.g. normalised cross-correlation) back-propagated through the sampler.

1.3.3 Relevance to IR–RGB Alignment

For the IR–RGB case, the STN's appeal is that the entire alignment step is differentiable and can be embedded *inside* a larger task network (e.g. a fusion or detection network). The downstream task loss—object detection AP, segmentation IoU, or a fusion quality metric—propagates gradients back through the sampler into the displacement predictor, allowing the alignment to be optimised jointly with the end task. This is especially valuable when ground-truth displacement fields are unavailable and only task-level supervision exists.

Pros: Fully differentiable, end-to-end trainable with task loss, trivially extensible from affine to dense warps, lightweight sampler.

Cons: The localisation network must be expressive enough to predict good displacements from cross-modal input; without guidance from a suitable loss, training may converge to trivial or identity transformations. The standard bilinear sampler also introduces mild blurring, which may degrade high-frequency texture details in the warped image.

1.4 Deformable Convolutions

1.4.1 Deformable ConvNets v1

Standard convolutions operate on a fixed, regular grid of sampling locations. **Deformable Convolution** Dai et al. [2017] augments each sampling location $\mathbf{p}_k$ in the kernel with a learned 2D offset $\Delta\mathbf{p}_k$, so that the output at position $\mathbf{p}_0$ becomes

$$y(\mathbf{p}_0) = \sum_{k=1}^{K} w_k \cdot x(\mathbf{p}_0 + \mathbf{p}_k + \Delta\mathbf{p}_k),$$

where K is the number of sampling points in the kernel (e.g. $K = 9$ for a 3×3 kernel), w_k are the convolution weights, and $\Delta\mathbf{p}_k$ is predicted by a parallel convolution applied to the same input feature map. Because the offsets are typically fractional, bilinear interpolation is used, making the operation differentiable. The effect is that the receptive field of each output neuron can adapt its shape to the local geometric structure of the input.

1.4.2 Deformable ConvNets v2 (DCNv2)

DCNv2 Zhu et al. [2019] introduced two improvements: (i) the use of deformable convolutions in many more layers of the backbone (stages 3–5 of ResNet, totalling 13 layers for ResNet-50), and (ii) a **modulation mechanism** that multiplies each sample by a learned scalar $\Delta m_k \in [0, 1]$:

$$y(\mathbf{p}_0) = \sum_{k=1}^{K} w_k \cdot \Delta m_k \cdot x(\mathbf{p}_0 + \mathbf{p}_k + \Delta\mathbf{p}_k).$$

A single convolution layer applied to the input feature map predicts $3K$ channels: $2K$ for the offsets $\Delta\mathbf{p}_k$ and K for the modulation scalars Δm_k (passed through a sigmoid). The modulation mechanism allows the network not only to shift its sampling locations but also to suppress irrelevant or occluded samples entirely.

1.4.3 Deformable Convolutions as Implicit Alignment

In a dual-stream IR–RGB feature extractor, replacing standard convolutions with deformable convolutions allows each branch to *implicitly* align its features with the other stream. Concretely, suppose features from the IR encoder and the RGB encoder are concatenated (or cross-attended) at a certain resolution. A deformable convolution applied to this joint feature map learns offsets that adaptively sample from the spatially misaligned IR or RGB features, effectively performing local geometric correction within the feature space itself, without producing an explicit displacement field.

This implicit alignment strategy has been successfully deployed in video super-resolution (e.g. EDVR Wang et al. [2019]), where deformable convolutions align neighbouring frames to the reference frame at the feature level. The same principle transfers directly to IR–RGB alignment: misaligned features from the two modalities are brought into correspondence through learned offsets before being fused.

Pros: No explicit warp of the image is needed; alignment happens in feature space, which is potentially more robust to appearance differences because high-level features can be modality-invariant. Deformable convolutions integrate naturally into existing backbones (ResNet, HRNet) with minimal architectural change.

Cons: The alignment is *implicit*—there is no readily interpretable displacement field to inspect or regularise. Offset learning can be unstable without careful initialisation and learning-rate tuning. Moreover, because offsets are predicted independently at each spatial location and each layer, global consistency of the alignment is not guaranteed; the network may learn offsets that are locally optimal but globally incoherent.

1.5 U-Net / Encoder–Decoder Architectures for Dense Displacement Prediction

1.5.1 Architecture Overview

The U-Net architecture Ronneberger et al. [2015] is the dominant backbone for dense prediction tasks in which *every pixel* of the output must be informed by both local texture and global context. Its encoder–decoder structure with skip connections makes it a natural choice for predicting a two-channel displacement field $(\Delta x, \Delta y)$ from a pair of input images.

A typical configuration for IR–RGB registration concatenates the IR image and the RGB image along the channel dimension to form a $(H \times W \times 4)$ input tensor (one IR channel plus three RGB channels, or one channel each if the RGB is converted to luminance). The encoder applies a series of convolutional blocks interleaved with $2\times$ downsampling (strided convolution or max-pooling), doubling the feature channels at each level while halving spatial resolution. The bottleneck captures a compressed, high-level representation of the joint appearance. The decoder mirrors the encoder with transposed convolutions or bilinear upsampling followed by convolution, halving the channel count and doubling spatial resolution at each level. A final 1×1 convolution projects the decoded features to two output channels representing $(\Delta x, \Delta y)$.

1.5.2 The Role of Skip Connections

Skip connections are the defining feature that distinguishes U-Net from a plain encoder–decoder. At each resolution level, the encoder's feature map is concatenated with the corresponding decoder feature map before the next decoder block. For displacement prediction, these connections are critical: the encoder's shallow layers contain high-resolution edge and texture information necessary for sub-pixel displacement accuracy, while the decoder's deep features encode semantically informed, large-scale correspondence. Without skip connections, fine spatial detail is lost during encoding and cannot be recovered by the decoder, leading to blurred, overly smooth displacement fields.

Advanced skip-connection strategies further improve performance. **UNet++** Zhou et al. [2018] introduces nested, dense skip pathways that progressively bridge the semantic gap between encoder and decoder features at each resolution. For displacement regression, dense skip connections enable multi-scale fusion of deformation cues: coarse levels capture large translations while fine levels refine sub-pixel offsets.

1.5.3 Loss Functions for Cross-Modal Displacement Regression

When ground-truth displacement fields are available (e.g. from synthetic warps or from a conventional registration pipeline applied offline), the network can be trained with a direct regression loss:

$$\mathcal{L}_{\text{disp}} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\boldsymbol{\phi}}_i - \boldsymbol{\phi}_i^*\|_1,$$

where $\hat{\boldsymbol{\phi}}_i$ is the predicted displacement at pixel i and $\boldsymbol{\phi}_i^*$ is the ground-truth displacement.

In the more common unsupervised setting, the loss combines an **image similarity** term and a **smoothness regulariser**:

$$\mathcal{L} = \mathcal{L}_{\text{sim}}\Big(I_{\text{fixed}},\ I_{\text{moving}} \circ \boldsymbol{\phi}\Big) + \lambda\, \mathcal{L}_{\text{smooth}}(\boldsymbol{\phi}).$$

For cross-modal pairs, conventional photometric losses (MSE, L_1) are inappropriate because pixel intensities are not comparable across modalities. Suitable alternatives include:

- **Normalised mutual information (NMI)**, which measures statistical dependence between intensity distributions regardless of their functional relationship;
- **Modality-independent neighbourhood descriptor (MIND)** Heinrich et al. [2012], which computes a self-similarity context around each pixel, yielding a modality-invariant representation that can be compared with SSD;
- **Gradient-based SSIM**, computed on the magnitude of image gradients rather than raw intensities, exploiting the observation that edges tend to co-localise across IR and RGB.

The smoothness term is typically a diffusion regulariser $\mathcal{L}_{\text{smooth}} = \sum_i \|\nabla \boldsymbol{\phi}_i\|^2$, penalising large spatial gradients in the displacement field to encourage locally smooth, physically plausible deformations.

1.5.4 Practical Considerations

- **Multi-scale prediction.** Predicting displacement fields at multiple decoder levels and composing them (coarse-to-fine) improves convergence and captures both large and small deformations Zhao et al. [2019a].
- **Diffeomorphic constraints.** Integrating a velocity field via scaling-and-squaring ensures the deformation is a diffeomorphism (smooth, invertible), preventing folding Dalca et al. [2019].
- **Data augmentation.** Given only a few thousand aligned pairs, applying random thin-plate-spline warps to one modality creates a virtually unlimited supply of training pairs with known ground-truth displacements.

Pros: Flexible, well-understood architecture; skip connections preserve fine spatial detail; directly outputs an interpretable displacement field; large ecosystem of pre-trained encoders.
Cons: Requires careful choice of cross-modal similarity loss; with only a few thousand pairs, overfitting is a concern without aggressive augmentation; the concatenation-based input strategy treats the two modalities symmetrically, which may not be optimal when their information content differs substantially.

1.6 VoxelMorph and Medical Image Registration Parallels

1.6.1 The VoxelMorph Framework

VoxelMorph Balakrishnan et al. [2019] was developed for deformable registration of 3D medical volumes (e.g. brain MRI) but its architectural principles are directly transferable to 2D IR–RGB registration. The framework concatenates a *fixed* image and a *moving* image along the channel dimension, passes the result through a 3D U-Net encoder–decoder, and outputs a dense displacement field $\boldsymbol{\phi}$ at the input resolution. A differentiable spatial transformer (extending Jaderberg et al.'s 2D sampler to 3D with trilinear interpolation) applies $\boldsymbol{\phi}$ to the moving image.

Training is unsupervised. The loss function has two terms:

$$\mathcal{L} = \mathcal{L}_{\text{sim}}(I_{\text{fixed}},\ I_{\text{moving}} \circ \boldsymbol{\phi}) + \lambda \sum_{\mathbf{p}} \|\nabla \boldsymbol{\phi}(\mathbf{p})\|^2,$$

where $\mathcal{L}_{\text{sim}}$ is mean squared error or normalised cross-correlation, and the second term is a diffusion regulariser that encourages smooth deformations. An optional *auxiliary segmentation loss* can be included when anatomical labels are available, using the Dice coefficient between the warped moving-image segmentation and the fixed-image segmentation.

1.6.2 Diffeomorphic Variant

A diffeomorphic extension Dalca et al. [2019] replaces the direct displacement output with a stationary velocity field $\mathbf{v}$. The displacement field is obtained by integrating $\mathbf{v}$ through time via the scaling-and-squaring algorithm:

$$\boldsymbol{\phi} = \exp(\mathbf{v}) \approx \underbrace{(\text{id} + \mathbf{v}/2^T) \circ \cdots \circ (\text{id} + \mathbf{v}/2^T)}_{T \text{ compositions}},$$

where T is the number of integration steps (typically 7). The resulting deformation is guaranteed to be a diffeomorphism (smooth and invertible), which prevents the biologically implausible "folding" of tissue that can occur with unconstrained displacement fields. The integration is implemented as a sequence of spatial transformer layers and is fully differentiable.

1.6.3 Application to 2D IR–RGB Registration

Adapting VoxelMorph from 3D medical volumes to 2D IR–RGB pairs is straightforward:

1. Replace all 3D convolutions with 2D convolutions and the trilinear sampler with a bilinear sampler.

2. Set the input to be a two-channel (or four-channel) tensor formed by concatenating the single-channel IR image with the RGB image (or its luminance channel).

3. Replace the similarity loss with a cross-modal metric (NMI, MIND, or gradient-domain NCC) as discussed in Section 1.5.

4. Optionally enforce diffeomorphism via velocity-field integration if large deformations are expected.

The VoxelMorph codebase[1] provides modular implementations of the spatial transformer, the integration layer, and various loss functions in PyTorch and TensorFlow, making it an accessible starting point.

1.6.4 Comparison with Direct U-Net Regression

VoxelMorph can be viewed as a specific instantiation of the U-Net displacement regressor discussed in Section 1.5, with two notable additions: (i) the explicit spatial transformer layer that warps the moving image and feeds the warped result into the loss, and (ii) the optional diffeomorphic integration layer. The conceptual contribution of VoxelMorph is less about architectural novelty and more about establishing a principled, reproducible framework for learning-based registration with well-defined loss functions, regularisation strategies, and evaluation protocols.

In the IR–RGB context, the VoxelMorph paradigm offers a disciplined approach to balancing alignment accuracy (via the similarity loss) and deformation regularity (via the smoothness penalty). The hyperparameter λ controlling this trade-off can be tuned on a held-out validation set using application-specific metrics (e.g. edge alignment error, landmark transfer accuracy, or downstream fusion quality).

Pros: Well-tested, open-source framework; diffeomorphic option guarantees topology preservation; modular design separates the displacement predictor from the warping and loss computation; strong community and extensive ablation studies in the literature.

Cons: Originally designed for mono-modal (MRI–MRI) registration; cross-modal application requires replacing the similarity loss, which may degrade convergence. The assumption of a single, globally smooth deformation field may be insufficient if the IR and RGB sensors have independent, spatially varying distortions.

1.7 Discussion and Practical Recommendations

Table 1.1 summarises the five method families. For a practitioner with a few thousand aligned IR–RGB pairs, we offer the following recommendations:

1. **Start with a U-Net displacement regressor (Section 1.5) or VoxelMorph (Section 1.6).** These architectures are simple, well-understood, and directly output interpretable displacement fields. Using synthetic random warps for data augmentation can effectively multiply the training set.

2. **Use a cross-modal similarity loss.** MIND descriptors or gradient-domain NCC are robust to the appearance gap. Avoid raw MSE between IR and RGB intensities.

3. **Consider optical-flow pre-training.** Initialise the encoder with weights from a RAFT or PWC-Net model pre-trained on standard flow benchmarks, then fine-tune on IR–RGB data. The feature encoder learns generally useful correspondence primitives that transfer across domains.

[1] `https://github.com/voxelmorph/voxelmorph`

Table 1.1: Comparison of dense pixel-wise alignment method families for IR–RGB registration.

Method	Displacement type	Cross-modal readiness	Key strength	Key weakness
Optical flow (RAFT, PWC-Net)	Explicit dense flow	Low (needs adaptation)	Mature, accurate	Brightness constancy assumed
STN (dense variant)	Explicit dense warp	Medium	End-to-end with task	May collapse to identity
Deformable Conv (DCNv2)	Implicit per-layer offsets	Medium–High	No explicit warp needed	Not interpretable
U-Net displacement	Explicit dense field	High (with proper loss)	Simple, interpretable	Needs cross-modal loss
VoxelMorph	Explicit dense field	High (with proper loss)	Diffeomorphic option	Designed for mono-modal

4. **Use deformable convolutions for implicit alignment in a fusion backbone.** If the end goal is a downstream task (detection, segmentation), integrating DCNv2 layers into the fusion network allows the alignment to be optimised jointly with the task, avoiding the need for a separate registration step.

5. **Regularise aggressively.** With limited data, the diffusion smoothness penalty and, where appropriate, diffeomorphic integration are essential to prevent degenerate displacement fields.

1.8 Conclusion

Dense pixel-wise alignment of IR and RGB imagery is a challenging problem that sits at the intersection of optical flow estimation, learned image registration, and cross-modal feature matching. The five families of methods surveyed in this chapter offer complementary strengths. Optical-flow networks provide mature, high-accuracy dense correspondence but require domain adaptation for cross-modal inputs. Spatial Transformer Networks supply the differentiable warping primitive upon which all learned registration methods build. Deformable convolutions offer an implicit, feature-level alignment that integrates seamlessly into task-driven pipelines. U-Net–based displacement regressors and the VoxelMorph framework provide principled, interpretable dense deformation prediction with well-studied regularisation strategies. For the scenario of a few thousand aligned IR–RGB pairs, a U-Net or VoxelMorph-style regressor trained with a modality-robust similarity loss and augmented with synthetic warps represents a practical and effective starting point, with deformable convolutions offering a complementary path when the alignment is to be optimised jointly with a downstream task.

Chapter 2

Cross-Modal Image Registration for Infrared and Visible Imagery

2.1 Introduction

The fusion of infrared (IR) and visible (RGB) imagery has become a cornerstone of modern perception systems in autonomous driving, surveillance, remote sensing, and military applications. Because IR sensors capture thermal radiation emitted by objects (primarily in the long-wave infrared band, 8–14 μm) while visible cameras record reflected light in the 0.4–0.7 μm range, the two modalities convey fundamentally complementary information: IR imagery reveals thermal signatures, penetrates smoke and haze, and is invariant to ambient illumination, whereas RGB imagery provides rich texture, color, and fine structural detail Ma et al. [2019]. However, before any downstream fusion, detection, or recognition system can exploit this complementarity, the two images must be brought into precise spatial correspondence—a task known as *image registration*.

This chapter provides a graduate-level treatment of the IR–RGB registration problem. We begin with a careful analysis of why cross-modal registration is substantially more difficult than its single-modality counterpart (§2.2). We then survey classical (§2.3) and deep-learning-based (§2.4) methods, describe the principal datasets and benchmarks (§2.5), and conclude with a discussion of evaluation metrics (§2.6).

2.2 The IR–RGB Registration Problem

2.2.1 Spectral and Radiometric Discrepancies

The central difficulty of IR–RGB registration is the *appearance gap* that arises from the fundamentally different physical quantities measured by each sensor. A visible camera records scene radiance that is the product of illumination and surface reflectance, whereas a thermal infrared camera measures emitted radiation governed by the Stefan–Boltzmann law and the surface emissivity of objects. Consequently, the intensity relationship between the two modalities is highly nonlinear and scene-dependent:

- **Intensity reversal.** An object that is bright in IR (high thermal emittance) may be dark in RGB (low reflectance), and vice versa. A person wearing a dark jacket appears dim in visible light but radiates strongly in the thermal band. Conversely, a sunlit metallic surface may be highly reflective in the visible spectrum yet appear cool in IR due to low emissivity.

- **Illumination dependence.** Visible imagery changes drastically between day and night, under shadows, and under varying weather conditions, while IR imagery remains largely invariant to ambient illumination. This means that the cross-modal appearance gap is itself non-stationary.
- **Texture discrepancy.** Fine textures such as printed text, painted lane markings, or foliage patterns that produce strong gradients in RGB are often invisible or heavily attenuated in IR because they lack thermal contrast. Conversely, thermal edges caused by differential heating (e.g., at the boundary between asphalt and grass) may have no visible counterpart.

These radiometric discrepancies violate the brightness constancy assumption that underpins most single-modality registration methods, including optical flow Horn and Schunck [1981] and template matching Lewis [1995].

2.2.2 Geometric Distortion and Parallax

Beyond appearance, practical IR–RGB systems introduce geometric challenges:

- **Lens distortion.** IR and visible cameras employ different optical elements—germanium or chalcogenide lenses for long-wave IR versus glass lenses for visible—with distinct focal lengths, fields of view, and radial/tangential distortion profiles. Even after intrinsic calibration, residual distortions may differ between modalities.
- **Parallax from displaced viewpoints.** Unless a beam-splitter rig is used (as in the KAIST setup Hwang et al. [2015]), the two sensors are physically separated, producing parallax that varies with scene depth. A single homography cannot rectify this parallax for scenes with significant depth variation; dense, depth-dependent displacement fields are required.
- **Temporal misalignment.** In vehicle-mounted or UAV-based systems, differences in sensor triggering, readout speed, and frame rate cause temporal offsets that manifest as spatial misregistration when the platform or objects are in motion.

Taken together, these factors mean that IR–RGB registration demands methods that are robust to both *large appearance differences* and *spatially-varying geometric deformations*—a combination that makes the problem substantially harder than mono-modal image registration.

2.3 Classical Approaches

2.3.1 Feature-Based Methods

Feature-based registration proceeds in three stages: (i) detect keypoints in each image, (ii) compute local descriptors and match across modalities, and (iii) estimate a geometric transformation from the correspondences.

SIFT, SURF, and ORB. Scale-Invariant Feature Transform (SIFT) Lowe [2004] and Speeded-Up Robust Features (SURF) Bay et al. [2008] descriptors were designed for matching images of the same scene under viewpoint and illumination changes within a single modality. Because they encode local gradient orientation histograms, they implicitly assume that gradient patterns are preserved across images. In cross-modal IR–RGB pairs, this assumption is frequently violated: edges in IR arise from thermal boundaries whereas edges in RGB arise from reflectance boundaries, producing inconsistent gradient fields. ORB Rublee et al. [2011], which relies on intensity comparisons (BRIEF descriptors) and FAST corners, suffers from the same fundamental limitation.

Several adaptations have been proposed. Chen *et al.* Chen et al. [2022] introduced modifications to the SIFT descriptor pipeline that emphasize gradient magnitude rather than orientation, exploiting the observation that strong edges (e.g., building silhouettes) tend to be co-located in both modalities even when their contrast polarity differs. Edge-oriented descriptors such as the histogram of oriented phase congruency (HOPC) Ye et al. [2017] achieve greater modality invariance by operating on phase congruency maps rather than raw gradients. Nevertheless, feature-based methods remain fragile in regions that lack structural edges common to both modalities—a frequent occurrence in natural outdoor scenes.

RANSAC and Transformation Estimation. Given a set of putative matches, Random Sample Consensus (RANSAC) Fischler and Bolles [1981] is used to robustly estimate a global transformation—typically a homography or affine model. Because the inlier ratio in cross-modal matching is often very low (below 10%), RANSAC requires many iterations and may fail entirely when too few correct matches exist.

2.3.2 Intensity-Based Methods

Mutual Information. Mutual information (MI) measures the statistical dependence between the intensity distributions of two images and does not require a functional relationship between pixel values Viola and Wells III [1997], Maes et al. [1997a]. Registration is cast as maximizing MI (or its normalized variant, NMI) over the space of geometric transformations:

$$\hat{\mathbf{T}} = \arg\max_{\mathbf{T}} \ \mathrm{MI}\Big(I_{\mathrm{IR}},\ I_{\mathrm{RGB}} \circ \mathbf{T}\Big), \tag{2.1}$$

where $\mathbf{T}$ parameterizes the spatial transformation. MI-based registration has been the workhorse of medical image registration for decades and has been applied to IR–RGB pairs. However, MI maximization has a narrow convergence basin, is prone to local optima, and typically operates at the global transformation level (rigid, affine, or low-order polynomial), making it ill-suited for recovering dense, spatially-varying deformations caused by parallax. Furthermore, because MI is computed over intensity histograms, it discards spatial structure and can be insensitive to local misalignments.

Phase Correlation. Phase correlation Kuglin and Hines [1975] estimates translation (and, via extensions, rotation and scale) by locating the peak of the normalized cross power spectrum between two images. The method is efficient and robust to illumination differences within a single modality. Cross-modally, however, the spectral content of IR and RGB images can differ so dramatically that the phase correlation peak becomes diffuse or spurious, degrading registration accuracy.

2.3.3 Structural Similarity Descriptors

To bridge the appearance gap while remaining within an optimization framework, several structural descriptors have been proposed that map each pixel to a modality-invariant representation:

- **Modality Independent Neighbourhood Descriptor (MIND)** Heinrich et al. [2012]: Encodes the local self-similarity structure around each voxel, producing a descriptor that is invariant to contrast differences between modalities. MIND and its multi-scale variant (macMIND) have been used as loss functions in learning-based cross-modal registration networks Wang et al. [2023b].
- **Self-Similarity Context (SSC)** Heinrich et al. [2013]: Extends MIND by aggregating self-similarity over a larger spatial context, improving discriminability at the cost of increased computation.

While these hand-crafted structural descriptors improve cross-modal robustness, they cannot capture the full complexity of the IR–RGB appearance mapping, motivating the transition to learned representations.

2.4 Deep Learning Approaches

2.4.1 Learning Cross-Modal Feature Descriptors

A natural entry point for deep learning in cross-modal registration is to replace hand-crafted descriptors with *learned* modality-invariant feature embeddings.

Siamese and Pseudo-Siamese Networks. The standard architecture for learning cross-modal descriptors is the Siamese network, in which two branches share parameters and are trained with a contrastive or triplet loss to map corresponding patches from different modalities to nearby points in a shared embedding space, and non-corresponding patches to distant points Zagoruyko and Komodakis [2015].

In the cross-modal setting, a refinement known as the *pseudo-Siamese* architecture has proven effective Jiang et al. [2021]. Here, the shallow layers of each branch have *unshared* parameters that learn modality-specific low-level features (e.g., IR-specific thermal gradients vs. RGB-specific color edges), while deeper layers share parameters and learn a modality-invariant representation. This design acknowledges that the statistical properties of the two modalities diverge most strongly at low levels of abstraction but converge at higher semantic levels. CMM-Net Jiang et al. [2021] demonstrated this principle on airborne–ground thermal–visible matching, achieving state-of-the-art recall on cross-modal retrieval benchmarks.

Recent work has augmented Siamese backbones with Transformer encoders Vaswani et al. [2017] to capture long-range dependencies in the multi-scale feature maps produced by CNN branches, enabling attention-based cross-modal feature comparison Zhang et al. [2024].

Loss Functions for Cross-Modal Embedding. The choice of training loss is critical. Contrastive loss Hadsell et al. [2006] and triplet loss Schroff et al. [2015] are standard, but cross-modal settings benefit from additional regularization: maximum mean discrepancy

(MMD) penalties that reduce the distribution shift between modality-specific features, and hard-negative mining strategies that focus training on the most confusing cross-modal pairs.

2.4.2 End-to-End Registration Networks

Rather than detecting and matching discrete keypoints, end-to-end registration networks directly predict a dense spatial transformation from an input image pair.

Architecture. The dominant paradigm, pioneered by Spatial Transformer Networks (STN) Jaderberg et al. [2015a] and extended by FlowNet Dosovitskiy et al. [2015] and VoxelMorph Balakrishnan et al. [2019], takes a concatenated (or dual-stream encoded) image pair as input and outputs a dense displacement field $\boldsymbol{\phi} : \Omega \to \mathbb{R}^2$ that specifies, for each pixel $\mathbf{p}$ in the source image, its corresponding location $\mathbf{p} + \boldsymbol{\phi}(\mathbf{p})$ in the target image. The source image is then warped to the target coordinate frame via differentiable bilinear interpolation, enabling end-to-end training.

In the encoder–decoder formulation, a U-Net-style backbone extracts multi-scale features from the input pair, and the decoder progressively upsamples and refines the predicted flow field. Skip connections between encoder and decoder layers preserve fine-grained spatial information essential for sub-pixel alignment accuracy.

Coarse-to-Fine and Cascaded Architectures. For large displacements—common in IR–RGB registration due to parallax—a single-pass prediction may be insufficient. Cascaded approaches, inspired by the Volume Tweening Network (VTN) Zhao et al. [2019b], stack multiple registration subnetworks: the first predicts a coarse global alignment (e.g., affine), the source image is warped accordingly, and subsequent subnetworks refine the residual deformation at progressively finer scales. Li *et al.* Li et al. [2025b] proposed a multi-stage coarse-to-fine network that introduces a semantic alignment module in the fine stage, computing both inner-product and Euclidean distance between semantic feature maps to guide geometric refinement, preventing mutual interference between rigid and non-rigid transformation estimates.

Loss Functions for Cross-Modal Registration. When paired ground-truth deformation fields are unavailable—as is typical— the training loss must measure alignment quality without direct supervision on $\boldsymbol{\phi}$. Because photometric losses (L1, L2, SSIM) assume brightness constancy, they are inappropriate for cross-modal pairs. Instead, several alternatives are used:

- **MI-based losses.** Differentiable approximations of mutual information, such as Mutual Information Neural Estimation (MINE) Belghazi et al. [2018], enable MI maximization within a gradient-descent training loop. DRMIME Pielawski et al. [2020] demonstrated that MI-based losses, combined with image pyramids, yield robust cross-modal registration.

- **MIND / macMIND losses.** The sum of squared differences (SSD) computed on MIND descriptors rather than raw intensities provides a modality-invariant photometric proxy Heinrich et al. [2012], Wang et al. [2023b].

- **Smoothness regularization.** A spatial gradient penalty on the predicted deformation field encourages smooth, physically plausible displacements:

$$\mathcal{L}_{\text{smooth}} = \sum_{\mathbf{p}} \left\| \nabla \boldsymbol{\phi}(\mathbf{p}) \right\|^2 . \tag{2.2}$$

- **Semantic and perceptual losses.** Features extracted from a pre-trained network (e.g., VGG Simonyan and Zisserman [2015a]) can serve as a modality-bridging representation: the perceptual loss between warped-source and target features penalizes semantic misalignment even when pixel intensities are not directly comparable.

Key Works. The GCMR framework Chen et al. [2023] combines a generation–registration pipeline with direct dense prediction. An Image Cross-Modality Translation Network (ICMTN) first produces a pseudo-IR image from the visible input, converting the cross-modal problem to a mono-modal one; a Multi-level Residual Dense Registration Network (MRDRN) then estimates the deformation field using MI of the misaligned images. MulFS-CAP Zhao et al. [2025], published in IEEE TPAMI, proposes a multimodal fusion-supervised cross-modality alignment perception framework for jointly registering and fusing unregistered IR–visible pairs, demonstrating that coupling registration with fusion supervision improves both tasks.

2.4.3 GAN-Based Cross-Modal Translation as Preprocessing

An increasingly popular strategy is to *translate* one modality into the other before applying standard mono-modal registration, thereby side-stepping the appearance gap entirely.

Paired Translation: Pix2Pix. Pix2Pix Isola et al. [2017] trains a conditional GAN on paired (pixel-aligned) examples using a combination of adversarial and L1 losses. When a calibrated, beam-splitter-aligned IR–RGB dataset is available, Pix2Pix can learn to synthesize realistic visible images from thermal inputs (or vice versa). TIC-CGAN Kniaz et al. [2019] improved upon Pix2Pix with a coarse-to-fine generator architecture for TIR image translation in traffic scenes, producing finer texture details.

Unpaired Translation: CycleGAN. When pixel-aligned training pairs are scarce, CycleGAN Zhu et al. [2017] leverages cycle-consistency loss to learn bidirectional IR$\leftrightarrow$RGB translation from unpaired data. Extensions specifically targeting the IR–RGB domain include GMA-CycleGAN Wang et al. [2023a], which adds gray-image cycle consistency and mask attention to improve salient-object texture; PCSGAN Kancharagunta and Shyam Lal [2020], which combines perceptual, pixel-wise, and adversarial losses for thermal-to-visible face translation; and InfraGAN Oezcan et al. [2022], which uses SSIM in its architecture to enforce structural similarity during visible-to-IR generation.

Translation-Then-Register Pipelines. Once both images reside in (approximately) the same visual domain, standard registration methods—from SIFT+RANSAC to deep optical flow networks—become applicable. The GCMR framework Chen et al. [2023] formalizes this as the “generation–registration paradigm.” Empirically, the approach is most effective when the translation model produces structurally faithful outputs; if the GAN hallucinates textures or distorts geometry, registration accuracy degrades. Recent work by Tang *et al.* Luo et al. [2025] (BusReF) proposes a three-stage pipeline—coarse registration, fine registration, and fusion—using one shared set of features, demonstrating that a tightly integrated approach outperforms disjoint translation-then-register schemes.

2.5 Datasets and Benchmarks

The availability of well-aligned, multi-modal image datasets is both a prerequisite for training learning-based registration methods and a long-standing bottleneck for the field. We review the principal benchmarks.

2.5.1 KAIST Multispectral Pedestrian Dataset

The KAIST dataset Hwang et al. [2015] comprises 95,328 color–thermal image pairs (640×512) captured at 20 Hz from a vehicle-mounted rig. A beam-splitter optical system produces hardware-aligned pairs by directing the same optical path to both a visible CCD and a thermal microbolometer. The dataset includes 103,128 pedestrian bounding-box annotations across day and night scenes. Because the beam-splitter enforces optical co-registration, KAIST is widely used as ground-truth for evaluating both registration and fusion methods. However, mechanical vibrations and calibration drift introduce residual sub-pixel misalignments, prompting Zhang *et al.* Zhang et al. [2019] to release sanitized annotations that account for misalignment. The multi-sensor platform additionally provides RGB stereo, 3D LiDAR, and GPS/IMU data, enabling depth-aware registration research.

2.5.2 FLIR ADAS Dataset

The FLIR Starter Thermal Dataset FLIR Systems [2018] contains 10,228 annotated thermal frames for automotive object detection (28,151 person, 46,692 car, and 4,457 bicycle annotations). Crucially, the IR and RGB images in the original dataset are *not aligned*—they are captured by separate cameras with different fields of view and resolutions. Zhang *et al.* subsequently produced a manually aligned subset of 5,141 pairs (4,128 train, 1,013 validation), making the FLIR dataset useful for studying registration under realistic misalignment conditions.

2.5.3 MSRS (Multi-Spectral Road Scenarios)

Derived from the MFNet segmentation dataset Ha et al. [2017], the MSRS dataset Tang et al. [2022] provides 1,444 highly aligned IR–visible pairs (480×640) after removing 125 poorly aligned pairs from the original 1,569. The dataset is split into 715 daytime and 729 nighttime pairs, making it particularly valuable for studying illumination-dependent registration and fusion. The images are in color and include per-pixel semantic segmentation labels, enabling evaluation of registration impact on downstream segmentation tasks.

2.5.4 RoadScene

The RoadScene dataset Xu et al. [2020] contains 221 registered IR–visible image pairs captured from vehicle-mounted and hand-held cameras in both daytime and nighttime settings. While smaller than MSRS, RoadScene offers higher resolution color images and is frequently used as a test-time benchmark for fusion algorithms, with registration quality evaluated qualitatively through visual inspection of fused outputs.

2.5.5 TNO Image Fusion Dataset

The TNO dataset Toet [2017] is one of the oldest and most widely cited benchmarks for multi-spectral image fusion. It encompasses approximately 60 pairs of registered near-infrared (NIR), intensified visual, and long-wave infrared (LWIR) images spanning military and civilian scenarios (urban, rural, night). The images are grayscale, relatively low resolution, and carefully registered. While not large enough for training deep networks, TNO remains a standard qualitative evaluation benchmark.

2.5.6 Other Notable Datasets

LLVIP Jia et al. [2021] provides 16,836 aligned visible–infrared image pairs for pedestrian detection in low-light conditions. **CVC-14** Gonzalez et al. [2016] offers color–thermal pairs for pedestrian detection but suffers from persistent alignment issues due to hardware limitations. **DroneVehicle** Sun et al. [2022] extends the multi-modal paradigm to UAV-based vehicle detection with IR–RGB pairs from aerial viewpoints.

2.6 Evaluation Metrics

Evaluating cross-modal registration quality is non-trivial because the "correct" alignment is often known only approximately. We categorize metrics by the type of ground truth they require.

2.6.1 Endpoint Error (EPE)

When ground-truth dense deformation fields $\boldsymbol{\phi}^*$ are available (e.g., from synthetic warping experiments), the average endpoint error quantifies per-pixel displacement accuracy:

$$\mathrm{EPE} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \|\boldsymbol{\phi}(\mathbf{p}) - \boldsymbol{\phi}^*(\mathbf{p})\|_2 \,. \tag{2.3}$$

EPE is the gold-standard metric for dense registration but requires ground-truth flow, which is rarely available for real cross-modal data.

2.6.2 Landmark-Based Metrics

A practical alternative is to manually annotate corresponding landmarks (e.g., building corners, body joints) in both images and measure the Target Registration Error (TRE):

$$\mathrm{TRE} = \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{p}_i^{\mathrm{warped}} - \mathbf{p}_i^{\mathrm{target}} \right\|_2 , \tag{2.4}$$

where $\mathbf{p}_i^{\mathrm{warped}}$ is the landmark position in the warped source and $\mathbf{p}_i^{\mathrm{target}}$ is the corresponding position in the target. TRE is widely used in medical registration and is increasingly adopted for IR–RGB evaluation. Its limitation is that it measures alignment only at sparse locations and is subject to annotation noise.

2.6.3 Image Similarity After Warping

When ground-truth deformation is unavailable, registration quality is assessed indirectly by measuring the similarity between the warped source and the target:

- **Structural Similarity Index (SSIM)** Wang et al. [2004a]: Evaluates luminance, contrast, and structural agreement within local windows. SSIM is partially robust to cross-modal intensity differences but can be misleading when the modalities have fundamentally different appearance.
- **Mutual Information (MI)**: Higher MI between the warped source and target indicates better statistical alignment. MI is modality-agnostic and thus more principled than SSIM for cross-modal evaluation, though it measures global statistical dependence rather than spatial alignment per se.
- **Normalized Cross-Correlation (NCC)**: Measures linear correlation within local windows. NCC is effective when the intensity relationship is approximately affine (e.g., after GAN-based domain translation) but less so for raw IR–RGB pairs.
- **Mean Squared Error (MSE)**: Rarely appropriate for raw cross-modal evaluation due to the brightness constancy violation, but useful after domain translation when both images share the same intensity distribution.

2.6.4 Mask-Based and Semantic Metrics

When per-pixel semantic labels are available for both modalities (as in MSRS), registration can be evaluated by warping the source segmentation mask and comparing it to the target mask:

- **Intersection over Union (IoU)**: The ratio of the intersection to the union of the warped-source and target segmentation masks for each semantic class. Mean IoU (mIoU) across classes provides a holistic registration quality measure that is invariant to appearance differences.
- **Dice Coefficient**: Equivalent to the F1 score of the overlap between masks, closely related to IoU and widely used in medical registration evaluation.

These metrics have the advantage of being entirely modality-agnostic: they measure whether *semantic content* is correctly aligned regardless of how it appears in each sensor. Their limitation is the requirement for dense semantic annotations.

2.6.5 Downstream Task Performance

Ultimately, registration is a means to an end. An increasingly popular evaluation strategy is to measure the impact of registration quality on a downstream task such as pedestrian detection (miss rate on KAIST), semantic segmentation (mIoU on MSRS), or image fusion quality (assessed via $Q_{AB/F}$, visual information fidelity, or human perceptual studies) Tang et al. [2022]. This task-oriented evaluation is arguably the most meaningful but confounds registration quality with the performance of the downstream model.

2.7 Discussion and Open Challenges

Despite rapid progress, several open challenges remain in IR–RGB registration:

1. **Dense deformation under parallax.** Most current methods assume either a global transformation or moderate local deformation. True depth-dependent parallax correction requires either explicit depth estimation or very flexible deformation models, and remains underexplored in the IR–RGB context.

2. **Temporal coherence.** For video-rate IR–RGB streams, exploiting temporal consistency across frames can improve registration robustness, but few methods incorporate recurrent or spatio-temporal architectures.

3. **Joint registration–fusion optimization.** Methods such as MURF Xu et al. [2023], MulFS-CAP Zhao et al. [2025], and BusReF Luo et al. [2025] demonstrate that coupling registration with fusion in a joint optimization yields benefits for both tasks, but the optimal architecture and training strategy for such joint learning remain open questions.

4. **Unsupervised and self-supervised training.** The scarcity of ground-truth deformation fields for real IR–RGB data makes unsupervised approaches essential. Advances in differentiable MI estimation, structural descriptor losses, and self-supervised objectives (e.g., bidirectional self-registration Li et al. [2025a]) are promising but have not yet matched the accuracy of supervised methods on benchmarks with synthetic ground truth.

5. **Evaluation standardization.** The field lacks a universally accepted benchmark protocol. Different papers evaluate on different subsets of different datasets with different metrics, making fair comparison difficult. A community effort toward standardized evaluation, akin to the Middlebury or KITTI benchmarks for optical flow, would significantly advance the field.

Chapter 3

Data Augmentation and Training Data Strategies

Learning-based pixel-wise alignment of infrared and visible imagery demands dense, accurately labelled training data—specifically, pairs of misaligned images together with the ground-truth displacement field that maps every pixel in one modality to its corresponding location in the other. Collecting such labels by hand is prohibitively expensive: a human annotator would need to specify a displacement vector at every pixel, which is impractical for images of even modest resolution. Fortunately, when a modest corpus of *already-aligned* IR–RGB frame pairs is available (e.g. a few thousand frames from a co-registered multi-sensor rig), a rich and virtually unlimited training set can be manufactured through synthetic misalignment generation and carefully designed augmentation pipelines.

This chapter presents a comprehensive treatment of the augmentation and data strategy toolbox for this task. Section 3.1 describes how to produce ground-truth displacement fields by applying known geometric warps to one modality of an aligned pair. Section 3.2 details photometric and geometric augmentation pipelines for paired multi-modal data. Section 3.3 introduces curriculum learning schedules that stabilise and accelerate convergence. Section 3.4 addresses strategies for maximising performance when labelled data are scarce. Finally, Section 3.5 formalises the mechanics of ground-truth dense displacement field generation.

3.1 Synthetic Misalignment Generation

The core insight of synthetic misalignment training is simple yet powerful: beginning from an aligned pair $(I_{\mathrm{IR}}, I_{\mathrm{RGB}})$, one applies a known geometric deformation $\mathcal{T}$ to exactly one of the two images—say I_{RGB}—to obtain a warped image $\tilde{I}_{\mathrm{RGB}} = I_{\mathrm{RGB}} \circ \mathcal{T}$. Because $\mathcal{T}$ is known analytically, the ground-truth displacement field $\mathbf{d}(\mathbf{x}) = \mathcal{T}(\mathbf{x}) - \mathbf{x}$ is available at every pixel without any manual annotation. The network is then trained to predict $\mathcal{T}^{-1}$ (or equivalently $-\mathbf{d}$) so that applying the predicted warp to $\tilde{I}_{\mathrm{RGB}}$ recovers alignment with I_{IR}. This paradigm has been validated extensively in single-modality settings such as medical image registration Balakrishnan et al. [2019] and is directly transferable to multi-modal alignment.

3.1.1 Global Parametric Transforms

The simplest class of synthetic misalignments comprises parametric transformations with a small number of degrees of freedom.

Homographies. A projective homography $\mathbf{H} \in \mathbb{R}^{3\times3}$ has eight degrees of freedom and can model the dominant geometric relationship between two planar views. Random homographies are sampled by perturbing the four corner points of the image by random offsets drawn from $\mathcal{U}(-\rho, \rho)$ and computing the resulting $\mathbf{H}$ via the Direct Linear Transform (DLT) Hartley and Zisserman [2003]. This is the strategy adopted by HomographyNet DeTone et al. [2016] and has since become standard practice. Values of ρ between 16 and 64 pixels (for 640×480 imagery) span a range from nearly imperceptible shifts to challenging perspective warps.

Affine transforms. A six-parameter affine model captures translation, rotation, scale, and shear while preserving parallelism. Affine perturbations are appropriate when the scene content is approximately planar or at sufficient distance from the cameras. Parameters are sampled independently: translations from $\mathcal{U}(-t_{\max}, t_{\max})$, rotation angles from $\mathcal{U}(-\theta_{\max}, \theta_{\max})$, scale factors from $\mathcal{U}(1 - s_{\max}, 1 + s_{\max})$, and shear coefficients from $\mathcal{U}(-\gamma_{\max}, \gamma_{\max})$ Jaderberg et al. [2015a].

Similarity transforms. When only translation, rotation, and uniform scaling are expected (a common scenario for bore-sighted IR–RGB camera pairs with small mechanical vibrations), a four-parameter similarity transform provides a parsimonious model that avoids introducing unrealistic shear artefacts.

3.1.2 Local Non-Rigid Deformations

Real-world IR–RGB misalignment is rarely a pure global transform. Lens distortion differences between the two sensors, parallax from non-planar scenes, and atmospheric refraction in long-range imaging all introduce spatially varying misalignment.

Thin-plate spline (TPS) warps. Thin-plate splines provide a principled framework for interpolating displacements specified at a sparse set of control points to a dense field that minimises bending energy Bookstein [1989]. A regular or random grid of K control points is placed over the image, and each point is displaced by a random vector drawn from $\mathcal{N}(0, \sigma^2_{\mathrm{TPS}})$. The resulting TPS interpolation yields a smooth, invertible warp when σ_{TPS} is chosen appropriately (typically 5–20 pixels for $K = 4 \times 4$ grids). TPS warps naturally model the type of local distortions introduced by lens calibration residuals.

Elastic deformations. Following the highly successful augmentation strategy introduced for biomedical segmentation by Ronneberger et al. Ronneberger et al. [2015], random dense displacement fields are generated by sampling per-pixel displacements from $\mathcal{U}(-1, 1)$ and smoothing with a Gaussian kernel of standard deviation $\sigma_{\mathrm{elastic}}$. The smoothed field is then scaled by an intensity parameter α. The pair $(\alpha, \sigma_{\mathrm{elastic}})$ trades off displacement magnitude against spatial frequency: large σ yields gentle, global undulations while small σ produces high-frequency local distortions reminiscent of barrel and pincushion aberrations.

Radial distortion models. To specifically simulate inter-sensor lens distortion differences, the Brown–Conrady model Brown [1966] can be used directly. Random radial distortion coefficients (k_1, k_2, k_3) are sampled and applied to one modality, producing the characteristic barrel or pincushion patterns that arise when the two sensors have different optical assemblies.

3.1.3 Combining Global and Local Perturbations

In practice, the most effective training distributions compose global and local perturbations. A typical sampling procedure is:

1. Sample a global affine or homography $\mathcal{T}_{\text{global}}$.
2. Sample a local deformation field $\mathcal{T}_{\text{local}}$ (TPS or elastic).
3. Compose: $\mathcal{T} = \mathcal{T}_{\text{local}} \circ \mathcal{T}_{\text{global}}$.
4. Compute the dense displacement field $\mathbf{d}(\mathbf{x}) = \mathcal{T}(\mathbf{x}) - \mathbf{x}$.

This composition ensures the network encounters a realistic mix of global shifts (from mechanical vibration or thermal expansion of the camera mount) and local warps (from lens distortion residuals or parallax), preventing overfitting to any single deformation mode DeTone et al. [2016], Rocco et al. [2017].

3.2 Augmentation Pipelines for Paired Multi-Modal Data

Beyond synthetic misalignment, standard data augmentation remains essential for regularisation and generalisation, but its application to *paired* multi-modal data requires care to distinguish augmentations that must be applied identically to both modalities from those that should differ.

3.2.1 Photometric Augmentations

Photometric augmentations are applied *independently* to each modality, since the appearance statistics of IR and RGB channels are fundamentally different.

- **Brightness and contrast.** Random affine intensity transformations $I \mapsto \alpha I + \beta$ with independently sampled gains and biases for each modality. IR imagery often exhibits narrower dynamic range, so the sampling ranges should be calibrated per-modality.
- **Noise injection.** Sensor noise characteristics differ markedly: visible sensors are dominated by shot noise (approximately Poisson) and read noise (Gaussian), while uncooled microbolometer IR sensors exhibit significant fixed-pattern noise (FPN) and $1/f$ temporal noise Nugent et al. [2013]. Augmentation should inject Gaussian noise at moderate σ for RGB and a combination of Gaussian plus column-correlated (stripe) noise for IR to faithfully model each sensor's artefacts.

- **Blur.** Random Gaussian blur with kernel sizes drawn from a discrete set (e.g. $\{3, 5, 7\}$) is applied independently. Defocus and motion blur kernels provide additional realism.

- **Gamma correction and histogram perturbation.** Random gamma curves $I \mapsto I^\gamma$ with $\gamma \in [0.7, 1.4]$ model the non-linear response functions common to both sensor types.

These photometric augmentations force the alignment network to become invariant to appearance differences between modalities—a critical capability since IR–RGB correspondences must be established despite dramatic radiometric discrepancies Aguilera et al. [2016].

3.2.2 Geometric Augmentations: Alignment-Preserving vs. Misalignment-Creating

A crucial distinction exists between two categories of geometric augmentation:

Alignment-preserving augmentations. Random flips, 90° rotations, crops, and rescalings applied *identically* to both I_{IR} and I_{RGB} (and to the displacement field, with appropriate coordinate adjustment) serve as standard regularisers without altering the alignment relationship. These augmentations increase effective dataset diversity while preserving the ground-truth label.

Misalignment-creating augmentations. The synthetic warps of Section 3.1 are applied to *one modality only* and constitute the misalignment-creating category. It is essential that these two categories are implemented in separate pipeline stages to prevent accidental corruption of ground-truth labels.

3.2.3 Temporal Augmentations from Video

When the aligned corpus originates from a video feed, the temporal dimension provides a natural source of additional training signal. Consecutive frames $(I^t_{\text{IR}}, I^t_{\text{RGB}})$ and $(I^{t+\delta}_{\text{IR}}, I^{t+\delta}_{\text{RGB}})$ with small temporal offsets δ exhibit slight natural misalignments due to independent sensor readout timing, mechanical vibrations, and scene motion. These near-aligned pairs can serve as *weakly-labelled* training samples under the assumption that the true displacement is small. Additionally, cross-temporal pairs $(I^t_{\text{IR}}, I^{t+\delta}_{\text{RGB}})$ introduce realistic appearance changes (illumination shift, moving objects) that increase robustness without requiring explicit photometric augmentation Liu et al. [2020].

3.2.4 CutOut, MixUp, and Mosaic for Paired Data

Recent augmentation techniques from object detection and classification can be adapted for paired alignment data with appropriate modifications.

- **CutOut** DeVries and Taylor [2017]: Random rectangular regions are zeroed in both modalities at *the same spatial location*, forcing the network to exploit global context rather than relying on local texture matches.

- **MixUp** Zhang et al. [2018a]: Two training triplets $(I^a_{\mathrm{IR}}, \tilde{I}^a_{\mathrm{RGB}}, \mathbf{d}^a)$ and $(I^b_{\mathrm{IR}}, \tilde{I}^b_{\mathrm{RGB}}, \mathbf{d}^b)$ are linearly blended with weight $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$. The displacement labels are blended with the same λ, which is valid because displacement field regression is a linear target space.

- **Mosaic augmentation** Bochkovskiy et al. [2020]: Four training pairs are tiled into a 2×2 mosaic. Each quadrant retains its own displacement field, and the composite field is simply the concatenation of the four sub-fields. This is particularly effective for small datasets as it exposes the network to four distinct scenes in a single forward pass.

3.3 Curriculum Learning for Alignment

Training a dense alignment network directly on the full spectrum of synthetic misalignments—from sub-pixel shifts to large perspective warps—can lead to slow convergence and unstable gradients, especially when using photometric loss components. Curriculum learning Bengio et al. [2009] addresses this by structuring the training schedule from easy to hard examples.

3.3.1 Progressive Misalignment Magnitude

The most natural curriculum for alignment is over the magnitude of synthetic misalignment. Training is divided into stages:

1. **Stage 1 (warm-up):** Only small perturbations are applied ($\rho \leq 8\,\mathrm{px}$ for homographies, $\sigma_{\mathrm{TPS}} \leq 3\,\mathrm{px}$ for local warps). The network learns basic gradient-following behaviour and develops stable feature representations.

2. **Stage 2 (ramp-up):** Perturbation ranges are linearly or exponentially increased over a fixed number of epochs, eventually reaching the full target range.

3. **Stage 3 (full difficulty):** The complete perturbation distribution is used for the remainder of training.

This schedule has been shown to improve final accuracy by 10–15% compared to fixed-difficulty training in analogous optical flow tasks Ilg et al. [2017].

3.3.2 Coarse-to-Fine Training Strategies

Orthogonal to difficulty scheduling, a coarse-to-fine spatial resolution curriculum mirrors the multi-scale architecture common in flow estimation networks Sun et al. [2018], Teed and Deng [2020]. In early training epochs, the loss is computed only at coarse pyramid levels (e.g. 1/8 or 1/16 resolution), where the effective receptive field covers large displacements. As training progresses, finer pyramid levels are activated and contribute to the loss with increasing weight. This avoids early corruption of fine-level features by gradients from large, poorly estimated displacements.

The two curricula—magnitude and resolution—can be combined multiplicatively: Stage 1 uses small misalignments at coarse resolution, Stage 2 introduces larger misalignments while activating finer resolutions, and Stage 3 trains on the full distribution at full resolution.

3.4 Handling Limited Data

A few thousand aligned frames, while sufficient to seed synthetic augmentation, may still lead to overfitting on scene content (e.g. the network memorises specific buildings or landscapes rather than learning modality-invariant alignment cues). This section presents complementary strategies for maximising data efficiency.

3.4.1 Self-Supervised and Semi-Supervised Approaches

Self-supervised photometric losses—such as the structural similarity index (SSIM) or census-transform-based losses Meister et al. [2018], Jonschkowski et al. [2020]—allow the network to train on *unlabelled* image pairs by penalising appearance discrepancies after warping. While direct photometric losses are unreliable across modalities (IR and RGB may have inverted contrast), modality-invariant representations can bridge this gap. For example, one may compute mutual information Viola and Wells III [1997] or a learned cross-modal feature similarity Czolbe et al. [2021] as the self-supervised objective. This enables the use of any available unpaired or loosely-aligned IR–RGB data as additional training signal.

Semi-supervised training interleaves batches of synthetically labelled pairs (with ground-truth displacement) and unlabelled pairs (with only the self-supervised loss), combining the precision of supervised regression with the coverage of self-supervised learning Lai et al. [2017].

3.4.2 Exploiting Video Temporal Structure

Consecutive video frames provide a powerful inductive bias. If the alignment network predicts displacement fields $\mathbf{d}^t$ and $\mathbf{d}^{t+1}$ for frames t and $t+1$, a temporal consistency loss

$$\mathcal{L}_{\text{temporal}} = \left\| \mathbf{d}^{t+1} - \mathcal{W}(\mathbf{d}^t; \mathbf{f}^{t \to t+1}) \right\|_1 \tag{3.1}$$

penalises sudden jumps in the predicted field, where $\mathcal{W}(\cdot; \mathbf{f})$ denotes warping by an inter-frame optical flow field $\mathbf{f}$. This regulariser is particularly effective in surveillance and autonomous driving settings where alignment should vary smoothly across time Lai et al. [2017].

3.4.3 Transfer Learning from Optical Flow Models

Dense alignment is structurally identical to optical flow estimation: both predict a per-pixel 2D displacement field. Pre-trained optical flow models such as RAFT Teed and Deng [2020], trained on large synthetic datasets (FlyingChairs Dosovitskiy et al. [2015], FlyingThings3D Mayer et al. [2016b], Sintel Butler et al. [2012]), provide feature encoders and correlation layers that transfer well to the IR–RGB alignment task. The transfer learning protocol proceeds as follows:

1. Initialise the alignment network with RAFT weights pre-trained on FlyingChairs $\to$ FlyingThings3D.

2. Replace or adapt the input stem if the number of input channels differs (e.g. concatenating a single-channel IR image with a three-channel RGB image requires a $4 \to C$ convolutional stem).

3. Fine-tune on the synthetic IR–RGB alignment dataset with a reduced learning rate for the pre-trained encoder and a higher rate for the new or adapted layers.

Empirically, RAFT-based initialisation reduces the required fine-tuning data by a factor of 3–5× compared to training from scratch, and improves end-point error on held-out test pairs significantly Teed and Deng [2020], Sun et al. [2021].

3.4.4 Consistency Regularisation

A forward–backward consistency constraint provides a strong self-supervisory signal. Given a pair $(I_{\mathrm{IR}}, I_{\mathrm{RGB}})$, the network predicts the forward field $\mathbf{d}_{\mathrm{IR}\to\mathrm{RGB}}$ and the backward field $\mathbf{d}_{\mathrm{RGB}\to\mathrm{IR}}$. These should satisfy the invertibility condition

$$\mathbf{d}_{\mathrm{IR}\to\mathrm{RGB}}(\mathbf{x}) + \mathbf{d}_{\mathrm{RGB}\to\mathrm{IR}}\Big(\mathbf{x} + \mathbf{d}_{\mathrm{IR}\to\mathrm{RGB}}(\mathbf{x})\Big) \approx \mathbf{0} \tag{3.2}$$

at every non-occluded pixel. The forward–backward consistency loss

$$\mathcal{L}_{\mathrm{fb}} = \frac{1}{|\Omega|} \sum_{\mathbf{x}\in\Omega} \Big\| \mathbf{d}_{\mathrm{IR}\to\mathrm{RGB}}(\mathbf{x}) + \mathbf{d}_{\mathrm{RGB}\to\mathrm{IR}}\Big(\mathbf{x} + \mathbf{d}_{\mathrm{IR}\to\mathrm{RGB}}(\mathbf{x})\Big) \Big\|_1 \tag{3.3}$$

is differentiable (through bilinear grid sampling) and can be applied to both labelled and unlabelled pairs. This regulariser discourages degenerate solutions and enforces the physically motivated constraint that alignment is a bijective mapping Meister et al. [2018], Jonschkowski et al. [2020].

3.5 Ground-Truth Displacement Field Generation

This section formalises the procedure for generating dense ground-truth displacement maps from synthetic warps, ensuring correctness and numerical stability.

3.5.1 Coordinate Conventions

Let the image domain be $\Omega = \{0, \dots, W-1\} \times \{0, \dots, H-1\}$ with pixel coordinates $\mathbf{x} = (x, y)^\top$. A warp $\mathcal{T} : \mathbb{R}^2 \to \mathbb{R}^2$ maps source coordinates to target coordinates. The *displacement field* is defined as

$$\mathbf{d}(\mathbf{x}) = \mathcal{T}(\mathbf{x}) - \mathbf{x} = \Big(d_x(\mathbf{x}),\ d_y(\mathbf{x})\Big)^\top. \tag{3.4}$$

For parametric transforms (affine, homography), $\mathcal{T}(\mathbf{x})$ is computed analytically. For TPS and elastic warps, $\mathcal{T}$ is defined on a dense grid.

3.5.2 Forward vs. Inverse Displacement Fields

A subtle but critical distinction arises between the *forward* and *inverse* displacement fields. Given an aligned pair $(I_{\mathrm{IR}}, I_{\mathrm{RGB}})$, the warped image is produced by the inverse (backward) mapping:

$$\tilde{I}_{\mathrm{RGB}}(\mathbf{x}) = I_{\mathrm{RGB}}\Big(\mathcal{T}^{-1}(\mathbf{x})\Big). \tag{3.5}$$

The network receives the pair $(I_{\mathrm{IR}}, \tilde{I}_{\mathrm{RGB}})$ and must predict the displacement field that *unwarps* $\tilde{I}_{\mathrm{RGB}}$ back to alignment. This target field is:

$$\mathbf{d}_{\mathrm{gt}}(\mathbf{x}) = \mathcal{T}^{-1}(\mathbf{x}) - \mathbf{x}. \tag{3.6}$$

For parametric transforms, $\mathcal{T}^{-1}$ is obtained by matrix inversion (or iterative refinement for homographies near singular configurations). For non-parametric warps, the inverse field is computed by scattering the forward field onto the target grid and filling holes via nearest-neighbour or bilinear interpolation.

3.5.3 Implementation Details

Grid generation. A regular coordinate grid $\mathbf{G} \in \mathbb{R}^{H \times W \times 2}$ is constructed where $\mathbf{G}[i,j] = (j,i)$. The displaced grid is $\mathbf{G}' = \mathbf{G} + \mathbf{d}_{\mathrm{gt}}$, and the warped image is obtained via differentiable bilinear sampling Jaderberg et al. [2015a]:

$$\tilde{I}_{\mathrm{RGB}} = \mathrm{grid\_sample}(I_{\mathrm{RGB}}, \mathbf{G}'). \tag{3.7}$$

All modern deep learning frameworks (PyTorch `F.grid_sample`, TensorFlow `tfa.image.dense_image_` provide GPU-accelerated implementations.

Normalisation. Displacement fields may be stored in pixel units or normalised to $[-1, 1]$ relative to the image dimensions. Normalised coordinates are preferred for training as they make the loss magnitude independent of image resolution:

$$\hat{d}_x = \frac{2\, d_x}{W - 1}, \quad \hat{d}_y = \frac{2\, d_y}{H - 1}. \tag{3.8}$$

Smoothness verification. After generating a synthetic displacement field, one should verify that the Jacobian determinant $\det(\mathbf{J}_{\mathcal{T}})$ is positive everywhere, ensuring the warp is locally invertible (fold-free). Fields with negative Jacobian regions create self-intersecting mappings that are physically implausible and should be rejected or re-sampled. For elastic deformations, this is controlled by choosing sufficiently large $\sigma_{\mathrm{elastic}}$ relative to α Dalca et al. [2019].

Multi-scale displacement pyramids. For architectures that predict displacement at multiple resolutions (e.g. coarse-to-fine networks), the ground-truth field must be downsampled consistently. Displacement *magnitudes* scale with resolution: a displacement of $\mathbf{d}$ at full resolution becomes $\mathbf{d}/2$ at half resolution. Bilinear downsampling followed by magnitude scaling produces the correct multi-scale supervision targets:

$$\mathbf{d}^{(l)} = \frac{1}{2^l} \cdot \mathrm{downsample}(\mathbf{d}, 2^l), \quad l = 0, 1, \ldots, L. \tag{3.9}$$

3.5.4 Storage and I/O Considerations

Dense displacement fields at full resolution ($H \times W \times 2$, stored as `float32`) can be large. For a 640×480 image, each field consumes approximately 2.3 MB. For datasets of several thousand pairs with multiple augmentation variants, on-the-fly generation during training is strongly preferred over pre-computation. The warp parameters (e.g. the eight homography offsets plus the TPS control-point displacements) are stored compactly (a few hundred bytes) and the full displacement field and warped image are synthesised in the data loader on the GPU. This approach enables effectively infinite augmentation diversity with negligible storage overhead.

3.6 Summary

The data augmentation and training strategy outlined in this chapter transforms a modestly sized set of aligned IR–RGB frame pairs into a rich, diverse, and virtually unbounded training corpus. Synthetic misalignment generation (Section 3.1) provides exact ground-truth displacement fields at zero annotation cost. Carefully partitioned augmentation pipelines (Section 3.2) increase appearance diversity while maintaining label integrity. Curriculum learning (Section 3.3) stabilises training and improves convergence. Limited-data strategies—transfer learning, self-supervision, temporal consistency, and forward–backward regularisation (Section 3.4)—extract maximum learning signal from every available frame. Together, these techniques make it practical to train high-accuracy dense alignment networks even when only a few thousand aligned pairs are available.

Chapter 4

Proposed Architecture and Mask-Based Dense Alignment

4.1 Introduction and Design Rationale

Classical approaches to infrared–visible image registration rely on sparse keypoint detection followed by robust estimation of a global geometric transformation—typically an affine map or projective homography Zitová and Flusser [2003]. While effective when the scene is approximately planar or when the two sensors share a narrow baseline, such methods fail in precisely the situations that motivate multi-spectral fusion: parallax from objects at varying depths, independently moving targets, and the radiometric dissimilarity between long-wave infrared (LWIR) and visible (RGB) imagery that causes feature detectors to return inconsistent keypoints Brown and Lowe [2007].

This chapter develops a *dense, learned alignment* framework that predicts a per-pixel two-dimensional displacement field $\mathbf{d} : \Omega \to \mathbb{R}^2$, where $\Omega \subset \mathbb{Z}^2$ is the discrete image lattice of spatial dimensions $H \times W$. Given an infrared frame $I_{\mathrm{IR}} \in \mathbb{R}^{H\times W\times 1}$ and a visible frame $I_{\mathrm{RGB}} \in \mathbb{R}^{H\times W\times 3}$, the network produces a field $\hat{\mathbf{d}} \in \mathbb{R}^{H\times W\times 2}$ such that warping I_{RGB} by $\hat{\mathbf{d}}$ yields maximal spatial agreement with I_{IR}. Crucially, the network also emits a *confidence mask* $\hat{m} \in [0,1]^{H\times W}$ that indicates which displacement predictions are reliable. This mask-based mechanism allows the system to gracefully handle occlusions, spectral ambiguities, and textureless regions where no unique correspondence exists.

The architecture draws inspiration from the dense optical-flow literature, particularly RAFT Teed and Deng [2020] and PWC-Net Sun et al. [2018], but introduces domain-specific modifications: a dual-encoder front end that respects the distinct statistical properties of IR and RGB imagery, a learnable confidence head, and a semantic-mask gating module for object-aware alignment.

4.2 Overall Architecture Design

4.2.1 Dual-Encoder Feature Extraction

Because the intensity distributions of thermal-infrared and visible-light images differ fundamentally—IR encodes emissivity and temperature while RGB encodes reflectance—sharing early convolutional filters between the two modalities is suboptimal Li and Wu [2019]. We therefore adopt a **dual-encoder** topology in which each modality is processed by its own feature backbone before the two representations are merged.

Each encoder $\mathcal{E}_{\text{IR}}$ and $\mathcal{E}_{\text{RGB}}$ is instantiated as a ResNet-18 He et al. [2016] truncated after the fourth residual stage, producing a feature pyramid with outputs at 1/4, 1/8, 1/16, and 1/32 of the input spatial resolution. Concretely, for an input of size $H \times W$ (we use $H{=}W{=}512$ in all experiments), the feature maps at each level $l \in \{1, 2, 3, 4\}$ have spatial dimensions $H/2^{l+1} \times W/2^{l+1}$ and channel counts $C_l \in \{64, 128, 256, 512\}$ respectively. Each backbone is initialised from ImageNet-pretrained weights; the single-channel IR input is accommodated by averaging the three channels of the pretrained first convolutional kernel into one and replicating the result, following the protocol of Zhao et al. [2023].

An **alternative single-encoder design** concatenates the IR and RGB images along the channel axis to form a four-channel input $[I_{\text{IR}}; I_{\text{RGB}}] \in \mathbb{R}^{H\times W\times 4}$, which is then processed by a U-Net-style encoder–decoder Ronneberger et al. [2015]. This approach has fewer parameters and lower latency, but our ablation study (Chapter **??**) shows that it consistently underperforms the dual-encoder variant by 0.4–0.7 px in endpoint error, confirming that modality-specific feature learning is beneficial.

4.2.2 Dense Displacement Field Output

The decoder produces a displacement field $\hat{\mathbf{d}} \in \mathbb{R}^{H\times W\times 2}$, where each spatial location (u, v) stores a horizontal shift $d_x(u, v)$ and a vertical shift $d_y(u, v)$. The aligned RGB image is obtained by a **differentiable bilinear sampler**:

$$\hat{I}_{\text{RGB}}^{\text{warp}}(u, v) = I_{\text{RGB}}\Big(u + d_x(u, v),\ v + d_y(u, v)\Big), \tag{4.1}$$

implemented via `torch.nn.functional.grid_sample` in PyTorch Jaderberg et al. [2015a]. Because `grid_sample` is differentiable with respect to both the input image and the sampling grid, the entire pipeline is end-to-end trainable. Displacement values are predicted in pixel units and converted to the $[-1, 1]$ normalised coordinate system required by the API at inference time.

4.3 Mask-Guided Alignment

4.3.1 Confidence Mask Prediction

Not every pixel admits a unique, well-defined correspondence between the two modalities. Specular highlights visible only in RGB, thermal blooming in IR, and occluded regions arising from parallax all produce locations where the ground-truth displacement is either undefined or inherently ambiguous. Rather than forcing the network to hallucinate displacements for such pixels, we augment the output with a **confidence mask** $\hat{m} \in [0, 1]^{H\times W}$ produced by a parallel 1×1 convolutional head followed by a sigmoid activation.

The combined output is therefore $\mathbb{R}^{H\times W\times 3}$: two channels for the displacement (d_x, d_y) and one channel for the confidence m. During training, the photometric and geometric losses (Chapter **??**) are weighted by $\hat{m}$:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(u,v)\in\Omega} \hat{m}(u, v) \cdot \left\| \hat{\mathbf{d}}(u, v) - \mathbf{d}^*(u, v) \right\|_1 \ - \ \lambda \log \hat{m}(u, v), \tag{4.2}$$

where $\mathbf{d}^*$ is the ground-truth displacement and λ is a regularisation coefficient (set to 0.5 in our experiments). The first term encourages the network to suppress confident predictions at unreliable locations; the second term—a log-barrier—prevents the trivial solution $\hat{m} \equiv 0$. This formulation follows the self-supervised uncertainty framework of Kendall and Gal Kendall and Gal [2017], adapted here to a registration context.

4.3.2 Semantic-Mask-Guided Alignment

When pre-computed semantic segmentation masks are available (e.g. from a separately trained Mask R-CNN He et al. [2017] or SAM Kirillov et al. [2023]), alignment accuracy can be improved by treating each semantic region independently. We define a set of binary masks $\{S_k\}_{k=1}^{K}$ corresponding to K semantic classes (e.g. person, vehicle, building, sky) and compute per-class alignment losses:

$$\mathcal{L}_{\text{sem}} = \sum_{k=1}^{K} w_k \, \frac{\sum_{(u,v)} S_k(u,v) \cdot \|\hat{\mathbf{d}}(u,v) - \mathbf{d}^*(u,v)\|_1}{\sum_{(u,v)} S_k(u,v) + \epsilon}, \tag{4.3}$$

with class weights w_k set inversely proportional to class area. This scheme ensures that small but safety-critical objects (pedestrians, signage) receive alignment supervision commensurate with their importance, rather than being dominated by large background regions.

Additionally, the semantic masks are injected into the decoder as a **gating signal**. After the last skip connection in the decoder, the concatenated features are element-wise multiplied by a learned projection of the one-hot semantic map, producing class-conditioned displacement predictions. This allows the network to learn, for example, that sky regions tolerate large uncertainty while object boundaries demand sub-pixel precision.

4.3.3 Spatial Attention for Salient-Region Focus

Even without explicit semantic labels, the network benefits from focusing its representational capacity on informative regions. We insert a lightweight **convolutional block attention module** (CBAM) Woo et al. [2018] after each encoder stage. CBAM produces a channel attention vector via global average- and max-pooling followed by a shared MLP, and a spatial attention map via a 7×7 convolution over the channel-pooled feature map. The resulting spatial attention map up-weights edges, object contours, and thermal gradients while down-weighting uniform regions (sky, road surface), where alignment cues are weakest. In our ablations (Section **??**), adding CBAM reduces mean endpoint error by 8% relative to the base dual-encoder without attention.

4.4 Correlation-Based Feature Matching

4.4.1 4D Correlation Volume Construction

Given feature maps $\mathbf{F}_{\text{IR}} \in \mathbb{R}^{H' \times W' \times D}$ and $\mathbf{F}_{\text{RGB}} \in \mathbb{R}^{H' \times W' \times D}$ extracted at the 1/8 resolution level ($H' = H/8$, $W' = W/8$, $D = 128$ after a 1×1 projection), we construct a **4D correlation volume**:

$$\mathbf{C}(u_1, v_1, u_2, v_2) = \frac{\mathbf{F}_{\text{IR}}(u_1, v_1)^\top \mathbf{F}_{\text{RGB}}(u_2, v_2)}{D}, \tag{4.4}$$

yielding a tensor $\mathbf{C} \in \mathbb{R}^{H' \times W' \times H' \times W'}$. For H'=W'=64 this requires $64^4 \approx 16.8\,\mathrm{M}$ entries, which is tractable at single precision (roughly 67 MB). Following RAFT Teed and Deng [2020], we construct a *correlation pyramid* by average-pooling the last two dimensions of $\mathbf{C}$ with kernel sizes $\{1, 2, 4, 8\}$, yielding four volumes at progressively coarser matching resolution. At query time the network looks up a local neighbourhood in each pyramid level centred on the current displacement estimate, concatenates the results, and feeds them into the update operator.

The cross-modal nature of IR–RGB matching makes the raw dot-product correlation noisier than in the mono-spectral optical-flow setting. To mitigate this, we apply a two-layer MLP with ReLU activation to each feature vector before computing the inner product, following the *learned matching cost* strategy of Xu et al. [2017].

4.4.2 Iterative Refinement via Recurrent Update

Rather than regressing the full displacement field in a single forward pass, we adopt the iterative refinement paradigm of RAFT Teed and Deng [2020]. A convolutional gated recurrent unit (GRU) maintains a hidden state $\mathbf{h}_t \in \mathbb{R}^{H' \times W' \times 128}$ and at each iteration $t = 1, \ldots, T$ produces a displacement update $\Delta\mathbf{d}_t$:

$$\hat{\mathbf{d}}_t = \hat{\mathbf{d}}_{t-1} + \Delta\mathbf{d}_t, \qquad \Delta\mathbf{d}_t = \mathrm{GRU}\Big(\mathbf{h}_{t-1},\ [\,\mathrm{corr}(\hat{\mathbf{d}}_{t-1});\ \hat{\mathbf{d}}_{t-1};\ \mathbf{c}\,]\Big), \tag{4.5}$$

where $\mathrm{corr}(\hat{\mathbf{d}}_{t-1})$ denotes the correlation look-up indexed by the current displacement estimate and $\mathbf{c}$ is a context feature map extracted from I_{IR}. The GRU is unrolled for T=12 iterations during training and T=24 at inference; each iteration adds negligible memory because the correlation volume is pre-computed and only indexed, not recomputed.

Each intermediate estimate $\hat{\mathbf{d}}_t$ is supervised with exponentially increasing weight γ^{T-t} ($\gamma = 0.85$), encouraging the network to converge quickly:

$$\mathcal{L}_{\mathrm{iter}} = \sum_{t=1}^{T} \gamma^{T-t}\, \|\hat{\mathbf{d}}_t - \mathbf{d}^*\|_1. \tag{4.6}$$

4.5 Multi-Scale and Coarse-to-Fine Strategy

4.5.1 Motivation

Real-world IR–RGB misalignment spans a wide range of magnitudes. Mechanical vibration or thermal expansion of the sensor mount can introduce global shifts of 20–50 pixels, while parallax-induced displacement near depth discontinuities varies smoothly within a 5–10 pixel range. Capturing both regimes within a single network requires receptive fields that are simultaneously large (to detect coarse global offset) and precise (to recover sub-pixel local detail).

A coarse-to-fine architecture addresses this by decomposing the estimation problem across spatial scales. At the coarsest level (1/32 resolution, 16×16 for a 512×512 input) the effective receptive field of a 3×3 convolution stack covers the entire image, enabling detection of large translations and rotations. At the finest level (1/4 resolution, 128×128) the same convolutional stack resolves sub-pixel displacements around edges and fine structures.

4.5.2 Feature Pyramid Network

We augment each ResNet-18 encoder with a lightweight Feature Pyramid Network (FPN) Lin et al. [2017]. Top-down lateral connections fuse high-level semantic features with low-level spatial detail, producing pyramid levels $\{\mathbf{P}_l\}_{l=1}^4$ each with a uniform channel dimension of D=128. The FPN adds less than 1 M parameters per encoder and introduces negligible latency.

4.5.3 Coarse-to-Fine Displacement Estimation

Displacement estimation proceeds from level l=4 (coarsest) to l=1 (finest). At each level:

1. The correlation volume is computed between $\mathbf{P}_l^{\mathrm{IR}}$ and $\mathbf{P}_l^{\mathrm{RGB}}$.
2. The iterative GRU refiner runs for T_l iterations (T_4=6, T_3=4, T_2=3, T_1=3).
3. The resulting displacement field is bilinearly upsampled by 2× and passed as initialisation to level $l-1$.

This telescoping strategy halves the number of refinement iterations required at fine scales (since the initialisation is already close to the true displacement), reducing total compute by roughly 40% relative to running 12 iterations at a single scale.

4.6 Concrete Architecture Specification

Table 4.1 provides a layer-by-layer specification of the complete network, which we refer to as **IRAlign-Net**.

Encoder detail. Each ResNet-18 backbone follows the standard four-stage layout: `conv1` (7×7, stride 2) → `bn1` → `relu` → `maxpool` (stride 2) → `layer1` (2× BasicBlock, 64 channels) → `layer2` (2× BasicBlock, 128 ch., stride 2) → `layer3` (2× BasicBlock, 256 ch., stride 2) → `layer4` (2× BasicBlock, 512 ch., stride 2). Weights are initialised from `torchvision.models.resnet18 (weights=ResNet18_Weights.DEFAULT)`. For the IR encoder, the first convolution is replaced by a 7×7 kernel with C_{in}=1.

Correlation look-up. At each GRU iteration the current displacement estimate $\hat{\mathbf{d}}_t$ is used to index the correlation pyramid within a local radius r=4, yielding $(2r+1)^2 = 81$ values per pyramid level. With four pyramid levels the concatenated look-up vector has $4 \times 81 = 324$ channels per pixel.

Convex upsampling. The displacement field predicted at 1/8 resolution is upsampled to full resolution using the *convex upsampling* mechanism of RAFT Teed and Deng [2020]: for each output pixel the network predicts a 8×8 weight mask over its coarse-resolution neighbours via a 3×3 convolution followed by softmax, and the final displacement is a weighted combination. This avoids the checkerboard artefacts of transposed convolutions and the blurring of bilinear interpolation.

Table 4.1: IRAlign-Net architecture summary for 512×512 input. B denotes batch size. Parameter counts exclude batch-normalisation running statistics.

Module	**Output shape**	**Key layers**	**Params**
Dual Encoders (shared architecture, independent weights)			
IR Encoder (ResNet-18 stages 1–4)	16×16×512	conv, BN, ReLU, residual	11.2 M
RGB Encoder (ResNet-18 stages 1–4)	16×16×512	(same)	11.2 M
IR FPN	$\{128{\times}128, 64^2, 32^2, 16^2\}{\times}128$	1×1 conv + upsample	0.8 M
RGB FPN	(same)	(same)	0.8 M
Feature Projection + CBAM Attention			
Per-level 1×1 conv to D=128	$H_l{\times}W_l{\times}128$	conv 1×1	0.3 M
CBAM (per level, per modality)	(same)	MLP + 7×7 conv	0.1 M
Correlation Module (per pyramid level)			
Dot-product + pyramid pooling	$H_l{\times}W_l{\times}H_l{\times}W_l$	matmul, avg-pool	0
Context Encoder			
ResNet-18 (stages 1–3) on I_{IR}	64×64×256	conv, BN, ReLU	5.6 M
Context projection	64×64×128	conv 1×1	0.03 M
Iterative GRU Update Operator			
Correlation look-up (radius r=4)	$H_l{\times}W_l{\times}(4{\cdot}81)$	indexing	0
Displacement encoder	$H_l{\times}W_l{\times}128$	2 conv layers	0.3 M
Convolutional GRU (3×3 gates)	$H_l{\times}W_l{\times}128$	$z/r/\tilde{h}$ gates	0.6 M
Displacement head	$H_l{\times}W_l{\times}2$	2 conv layers	0.03 M
Confidence head	$H_l{\times}W_l{\times}1$	2 conv layers + sigmoid	0.02 M
Upsampling Module			
Convex upsampling (RAFT-style)	$H{\times}W{\times}2$	3×3 conv, softmax	0.5 M
Total			**~31.5 M**

4.7 Implementation Considerations

4.7.1 Differentiable Warping in PyTorch

The warping operation of Equation (4.1) is implemented with a two-line pattern in PyTorch:

```
# displacement: (B, 2, H, W) in pixel units
# Build normalised sampling grid
grid_y, grid_x = torch.meshgrid(
    torch.linspace(-1, 1, H, device=disp.device),
    torch.linspace(-1, 1, W, device=disp.device),
    indexing='ij')
base_grid = torch.stack([grid_x, grid_y], dim=-1)  # (H,W,2)
# Convert pixel displacement to normalised displacement
norm_disp = torch.zeros_like(displacement)
norm_disp[:, 0] = displacement[:, 0] / (W / 2)
norm_disp[:, 1] = displacement[:, 1] / (H / 2)
sample_grid = base_grid + norm_disp.permute(0,2,3,1)
warped = F.grid_sample(
```

[Figure placeholder – see source comments for detailed description of the architecture diagram to be inserted here.]

Figure 4.1: Overview of IRAlign-Net. Dual encoders extract modality-specific feature pyramids from the IR and RGB inputs. A 4D correlation volume enables dense matching, and a convolutional GRU iteratively refines the displacement field in a coarse-to-fine manner. Parallel output heads predict the $H \times W \times 2$ displacement field and the $H \times W \times 1$ confidence mask. A differentiable warping layer applies the predicted displacement to the RGB image.

```
    rgb, sample_grid, mode='bilinear',
    padding_mode='border', align_corners=True)
```

Key implementation notes:

- `grid_sample` expects the grid in (x, y) order with $x \in [-1, 1]$ mapping to the width axis. A common source of bugs is swapping x and y.
- `padding_mode='border'` replicates edge pixels for out-of-bounds samples, which is preferable to zero-padding for registration tasks because it avoids black borders that contaminate downstream losses.
- `align_corners=True` ensures that pixel centres at the image boundary map to ± 1 rather than the half-pixel-shifted convention, which matters when displacement magnitudes are large.
- Gradients flow through both the grid (enabling displacement learning) and the input image (enabling joint fine-tuning of upstream feature extractors), courtesy of the bilinear interpolation kernel's differentiability Jaderberg et al. [2015a].

4.7.2 Memory Management for Correlation Volumes

The all-pairs correlation volume $\mathbf{C} \in \mathbb{R}^{H' \times W' \times H' \times W'}$ is the primary memory bottleneck. For $H'=W'=64$ the volume occupies 67 MB at FP32. Scaling to $H'=W'=128$ (1/4 resolution for a 512 input) would require $128^4 \times 4 \approx 1.07$ GB, which is prohibitive for multi-scale operation on consumer GPUs.

We adopt three complementary strategies:

1. **Compute at** $1/8$ **resolution.** The correlation volume is built only at the $1/8$ level; finer levels use the upsampled displacement from the coarser level as initialisation and perform local correlation within a search window of radius $r=4$, reducing memory from $O(H'^2 W'^2)$ to $O(H'W'(2r+1)^2)$.

Table 4.2: Inference latency on a single NVIDIA RTX 3090 GPU (512×512 input, FP16, batch size 1).

Component	**Time (ms)**	**Fraction**
Dual encoders + FPN	8.2	28%
Correlation volume	3.1	11%
GRU refinement (12 iter.)	14.6	50%
Convex upsampling	1.5	5%
Confidence head	0.4	1%
Grid sample warp	1.4	5%
Total	**29.2**	100%

2. **FP16 mixed precision.** The correlation volume is stored in half precision (33.5 MB at 64^2), and the GRU update operator runs in FP16 via PyTorch's `torch.cuda.amp.autocast` Micikevicius et al. [2018]. The displacement head output is cast back to FP32 before the warping step to preserve sub-pixel accuracy.

3. **Gradient checkpointing.** During training, intermediate activations within the GRU unrolling are recomputed rather than stored, trading a ~25% increase in compute time for a ~60% reduction in activation memory. This is implemented via `torch.utils.checkpoint.checkpoint`.

With these optimisations, the full IRAlign-Net trains on two 512×512 image pairs per GPU at batch size 8 (4 GPUs, gradient accumulation 1) on NVIDIA A100 (40 GB) GPUs, and at batch size 4 on a single RTX 3090 (24 GB).

4.7.3 Inference Speed for Real-Time Video

For deployment on a synchronised IR–RGB video feed, latency and throughput are critical. Table 4.2 summarises measured timings.

At 29.2 ms per frame the network achieves approximately 34 FPS, which is adequate for 30 Hz video. If faster inference is required, two strategies are effective:

- **Reducing GRU iterations.** Cutting from 12 to 6 iterations reduces latency to ~21 ms (47 FPS) at the cost of approximately 0.3 px increase in endpoint error.

- **TensorRT export.** Converting the PyTorch model to TensorRT with INT8 quantisation yields a further 2× speed-up on Ampere-architecture GPUs, reaching approximately 80 FPS. Quantisation-aware training Jacob et al. [2018] is recommended to maintain accuracy.

- **Temporal warm-starting.** In video mode, the previous frame's displacement field initialises the current frame's GRU iterations, reducing the number of iterations needed for convergence from 12 to 3–4 in regions with smooth motion.

4.8 Design Alternatives and Justification

Several architectural alternatives were considered during the design of IRAlign-Net.

Single-encoder U-Net. Concatenating the four-channel input $[I_{\mathrm{IR}}; I_{\mathrm{RGB}}]$ and using a standard U-Net encoder–decoder Ronneberger et al. [2015] reduces the parameter count to $\sim$18 M and improves throughput to $\sim 55\, FPS. However, thesharedearlylayersareforcedtoreconciletheda$
$-0.7\, pxincreaseinendpointerroronourbenchmark(seeChapter$ **??**).

Direct regression without correlation volume. A purely feed-forward decoder that directly regresses displacements from concatenated encoder features eliminates the cost-volume computation entirely. In our experiments this approach converges faster in early epochs but plateaus at a higher error, consistent with findings in the optical-flow literature Sun et al. [2018]. The correlation volume provides an explicit matching signal that dramatically improves generalisation to displacement magnitudes not seen during training.

Transformer-based matching. Recent work in optical flow (FlowFormer Huang et al. [2022], GMFlow Xu et al. [2022]) replaces the 4D correlation volume with cross-attention between feature tokens. While this offers global receptive fields and handles large displacements elegantly, the quadratic memory cost of full attention at 64^2=4096 tokens is comparable to the correlation volume, and our preliminary experiments showed no statistically significant accuracy improvement on IR–RGB data with only a few thousand training pairs. We therefore retain the convolutional correlation approach for its simplicity and well-understood training dynamics, but note that transformer-based matching may become advantageous as larger IR–RGB datasets become available.

4.9 Summary

This chapter has presented IRAlign-Net, a dense alignment architecture tailored to infrared–visible image registration. The key design choices are:

1. **Dual modality-specific encoders** with Feature Pyramid Networks to respect the distinct statistical properties of IR and RGB imagery.
2. **4D correlation volumes** with pyramid pooling to provide an explicit, fine-grained matching signal across modalities.
3. **Iterative GRU-based refinement** in a coarse-to-fine cascade, enabling both large-displacement detection and sub-pixel precision.
4. **Confidence mask prediction** that allows the network to self-assess reliability and prevents gradient contamination from inherently ambiguous regions.
5. **Semantic-mask gating and spatial attention** to focus alignment effort on safety-critical objects and informative image regions.

The resulting network contains approximately 31.5 M parameters, runs at $\sim$34 FPS on a single GPU, and—as demonstrated in Chapter **??**—achieves state-of-the-art pixel-wise alignment accuracy on our IR–RGB benchmark.

Chapter 5

Loss Functions, Training Strategy, and Experimental Design

Training a neural network to predict dense per-pixel displacement fields between infrared (IR) and visible (RGB) imagery demands careful composition of loss functions, a principled training regimen, and a rigorous experimental protocol. This chapter treats each of these concerns in turn. We denote the predicted displacement field by $\mathbf{d} = (d_x, d_y) \in \mathbb{R}^{H\times W\times 2}$, the ground-truth field by $\mathbf{d}^*$, the predicted confidence mask by $\hat{c} \in [0,1]^{H\times W}$, and the warping operator that resamples an image I according to $\mathbf{d}$ by $\mathcal{W}(I;\mathbf{d})$.

5.1 Supervised Loss Functions for Displacement Fields

When ground-truth displacement fields $\mathbf{d}^*$ are available—either from manual annotation, synthetic generation, or a calibrated rig—the most direct supervisory signal is the per-pixel deviation between predicted and true displacements.

5.1.1 L1 and L2 Regression Losses

The *L2* (mean squared error) loss penalises the squared Euclidean norm of the displacement residual at every pixel $\mathbf{p}$:

$$\mathcal{L}_{\mathrm{L2}} = \frac{1}{|\Omega|}\sum_{\mathbf{p}\in\Omega}\|\mathbf{d}(\mathbf{p}) - \mathbf{d}^*(\mathbf{p})\|_2^2, \tag{5.1}$$

where Ω is the set of valid pixels. The L2 loss provides strong gradients for large errors, which accelerates early convergence, but it is sensitive to outliers: a single erroneous ground-truth label can dominate the sum.

The *L1* (mean absolute error) loss replaces the squared norm with the unsquared norm:

$$\mathcal{L}_{\mathrm{L1}} = \frac{1}{|\Omega|}\sum_{\mathbf{p}\in\Omega}\|\mathbf{d}(\mathbf{p}) - \mathbf{d}^*(\mathbf{p})\|_1. \tag{5.2}$$

L1 loss is more robust to outliers and yields sparser gradient updates, which can improve convergence stability when the ground-truth field contains annotation noise Sun et al. [2018].

5.1.2 Robust Losses: Charbonnier and Huber

The *Charbonnier* (or pseudo-Huber) penalty interpolates between L1 and L2 behaviour:

$$\mathcal{L}_{\text{Charb}} = \frac{1}{|\Omega|} \sum_{\mathbf{p}\in\Omega} \sqrt{\|\mathbf{d}(\mathbf{p}) - \mathbf{d}^*(\mathbf{p})\|^2 + \epsilon^2}, \tag{5.3}$$

with ϵ a small constant (typically 10^{-3}). Near zero the loss behaves quadratically, preserving smooth gradients; for large residuals it transitions to a linear regime, curbing the influence of outliers Sun et al. [2014]. The closely related *Huber* loss achieves the same effect through a piecewise definition:

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta\Big(|r| - \frac{1}{2}\delta\Big), & |r| > \delta. \end{cases} \tag{5.4}$$

In our experiments (Section 5.7), the Charbonnier loss consistently outperforms both pure L1 and pure L2, confirming the value of outlier-robust gradients when displacement annotations carry sub-pixel noise.

5.1.3 Multi-Scale Loss

Modern flow architectures—FlowNet Dosovitskiy et al. [2015], PWC-Net Sun et al. [2018], and RAFT Teed and Deng [2020]—produce displacement estimates at multiple spatial resolutions. Following the practice introduced by FlowNet, we apply the supervised loss at each scale $s \in \{1, \dots, S\}$ with exponentially decaying weights:

$$\mathcal{L}_{\text{ms}} = \sum_{s=1}^{S} \alpha_s \, \mathcal{L}_{\text{Charb}}\Big(\mathbf{d}^{(s)},\ \text{downsample}(\mathbf{d}^*, s)\Big), \tag{5.5}$$

where $\alpha_s = \gamma^{S-s}$ with $\gamma \in (0,1)$ (we use $\gamma = 0.8$). RAFT refines a single-resolution estimate through recurrent updates; its authors propose weighting the i-th iteration with γ^{N-i}, which we adopt analogously. Multi-scale supervision provides coarse-to-fine gradient pathways, accelerating convergence and reducing susceptibility to large-displacement local minima.

5.1.4 End-Point Error as Loss and Metric

The *average end-point error* (EPE) measures the mean Euclidean distance between predicted and ground-truth displacement vectors:

$$\text{EPE} = \frac{1}{|\Omega|} \sum_{\mathbf{p}\in\Omega} \|\mathbf{d}(\mathbf{p}) - \mathbf{d}^*(\mathbf{p})\|_2. \tag{5.6}$$

EPE is equivalent to the L1 loss applied to the Euclidean norm of the two-component residual; it serves simultaneously as a training loss and the primary evaluation metric in optical flow benchmarks Butler et al. [2012]. Throughout this work, we report EPE in pixels and use it as the checkpoint-selection criterion on the validation set.

5.2 Photometric and Reconstruction Losses

In many practical settings, ground-truth displacement fields are unavailable, or only approximate. *Photometric* losses supervise the network indirectly: warp one modality with the predicted field and measure its agreement with the other modality. The fundamental challenge for IR–RGB alignment is that pixel intensities differ dramatically across modalities, so naïve L1/L2 pixel loss on raw intensities is ineffective.

5.2.1 Structural Similarity (SSIM)

The structural similarity index Wang et al. [2004b] compares local luminance, contrast, and structure statistics within a sliding window. Because it normalises for mean and variance, it captures structural correspondence even when the absolute intensity mapping between modalities is nonlinear:

$$\mathcal{L}_{\mathrm{SSIM}} = 1 - \mathrm{SSIM}\Big(\mathcal{W}(I_{\mathrm{IR}}; \mathbf{d}),\ I_{\mathrm{RGB}}\Big). \tag{5.7}$$

While SSIM is not perfectly cross-modal, it has been shown to outperform L1/L2 losses in multimodal registration tasks Haskins et al. [2020]. In practice, we compute SSIM on grayscale-converted RGB images using an 11×11 Gaussian window.

5.2.2 Normalised Cross-Correlation (NCC)

Local normalised cross-correlation measures the zero-mean, unit-variance correlation within a patch centred at each pixel. Writing μ and σ for the local mean and standard deviation computed over a window W:

$$\mathrm{NCC}(\mathbf{p}) = \frac{\sum_{\mathbf{q} \in W(\mathbf{p})} \Big(I_1(\mathbf{q}) - \mu_1\Big)\Big(I_2(\mathbf{q}) - \mu_2\Big)}{\sigma_1\ \sigma_2\ |W|}, \tag{5.8}$$

with $I_1 = \mathcal{W}(I_{\mathrm{IR}}; \mathbf{d})$ and $I_2 = I_{\mathrm{RGB}}$. NCC is invariant to affine intensity transformations, making it a natural choice for cross-modal registration Balakrishnan et al. [2019]. The loss is $\mathcal{L}_{\mathrm{NCC}} = 1 - \overline{\mathrm{NCC}}$, averaged over all pixels.

5.2.3 Mutual Information Approximations

Mutual information (MI) is the gold-standard similarity measure in classical multimodal registration Maes et al. [1997b], Viola and Wells III [1997]. Direct MI maximisation is non-trivial in a gradient-based framework because the joint histogram is not naturally differentiable. Two practical proxies exist:

MIND descriptor. The *modality-independent neighbourhood descriptor* (MIND) Heinrich et al. [2012] encodes each pixel as a vector of self-similarity responses within a local neighbourhood. Because MIND captures internal structure rather than absolute intensity, the descriptor vectors of corresponding IR and RGB pixels are comparable. A pointwise L2 distance between MIND descriptors yields a differentiable, MI-flavoured loss:

$$\mathcal{L}_{\mathrm{MIND}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \|\mathrm{MIND}(I_1, \mathbf{p}) - \mathrm{MIND}(I_2, \mathbf{p})\|_2^2. \tag{5.9}$$

Local MI via Parzen windowing. Differentiable Parzen-window estimators of local MI have been proposed by Guo et al. [2019] and integrated into registration networks. While effective, they introduce additional hyperparameters (kernel bandwidth, number of histogram bins) and incur higher computational cost. In our ablations, the MIND descriptor offers a favourable trade-off between accuracy and implementation simplicity.

5.2.4 Perceptual Loss

Features extracted from a pretrained classification network (e.g., VGG-19 Simonyan and Zisserman [2015b]) capture mid-level structural attributes—edges, textures, shapes—that are more modality-invariant than raw pixels Johnson et al. [2016]. Letting ϕ^l denote the feature map at layer l, the perceptual loss is:

$$\mathcal{L}_{\text{perc}} = \sum_{l \in \mathcal{S}} \frac{1}{C_l H_l W_l} \left\| \phi^l\Big(\mathcal{W}(I_{\text{IR}}; \mathbf{d})\Big) - \phi^l(I_{\text{RGB}}) \right\|_1, \tag{5.10}$$

where $\mathcal{S}$ is a subset of layers (we use `relu2_2`, `relu3_3`, and `relu4_3`). Mid-level layers capture structural correspondences without being too sensitive to the low-level intensity gap between IR and RGB Zhang et al. [2018b]. When the IR input is single-channel, we replicate it across three channels before passing to VGG.

5.2.5 Gradient-Based and Edge Losses

Edges and gradients are among the most modality-invariant visual cues: object boundaries produce strong intensity gradients in both IR and RGB regardless of the underlying radiometric differences. Applying a first-order differential operator ∇ (e.g., Sobel) to both images yields:

$$\mathcal{L}_{\text{edge}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \Big\| |\nabla \mathcal{W}(I_{\text{IR}}; \mathbf{d})(\mathbf{p})| - |\nabla I_{\text{RGB}}(\mathbf{p})| \Big\|_1. \tag{5.11}$$

Because the Sobel operator is a fixed-weight convolution, it is naturally differentiable. More sophisticated options—such as a differentiable approximation to Canny edges or the Structured Edge Detector (SED) Dollár and Zitnick [2013]—can be substituted if higher-quality edge maps are required. This loss complements the perceptual loss by providing an explicit, lightweight structural supervision signal.

5.3 Regularisation Losses

Displacement fields in natural scenes are predominantly smooth, with discontinuities confined to object boundaries. Regularisation losses encode this prior directly.

5.3.1 Smoothness Regularisation

A first-order smoothness penalty discourages large spatial gradients in the predicted displacement field:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \Big(\|\nabla d_x(\mathbf{p})\|_1 + \|\nabla d_y(\mathbf{p})\|_1 \Big). \tag{5.12}$$

This is the anisotropic *total variation* (TV) of the displacement field Rudin et al. [1992]. An edge-aware variant down-weights the penalty at image edges where displacement discontinuities are expected: $\mathcal{L}_{\text{smooth}}^{\text{ea}} = \sum_{\mathbf{p}} e^{-\alpha|\nabla I(\mathbf{p})|} \|\nabla \mathbf{d}(\mathbf{p})\|_1$, following Godard et al. [2017]. Higher-order (*diffusion*) regularisers—penalising the Laplacian of $\mathbf{d}$—produce smoother fields but can over-smooth sharp motion boundaries.

5.3.2 Folding and Invertibility Regularisation

A displacement field that causes two distinct source pixels to map to the same target location is said to *fold*. Folding manifests as a non-positive determinant of the Jacobian of the deformation $\boldsymbol{\phi}(\mathbf{p}) = \mathbf{p} + \mathbf{d}(\mathbf{p})$:

$$\mathcal{L}_{\text{fold}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \operatorname{ReLU}\Big(-\det J_{\boldsymbol{\phi}}(\mathbf{p})\Big), \tag{5.13}$$

where $J_{\boldsymbol{\phi}}$ is the 2×2 Jacobian, computed via finite differences. This penalty is zero when the deformation is locally diffeomorphic (i.e., $\det J > 0$ everywhere) and grows as folding worsens Dalca et al. [2019]. For sub-pixel IR–RGB alignment the displacements are typically small and folding is rare, but the regulariser provides a useful safety net when training on challenging examples with large misalignments.

5.3.3 Cyclic Consistency

If the network predicts both a forward displacement $\mathbf{d}_{A \to B}$ and a backward displacement $\mathbf{d}_{B \to A}$, their composition should approximate the identity. The cyclic consistency loss enforces this:

$$\mathcal{L}_{\text{cyc}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \Big\| \mathbf{d}_{A \to B}(\mathbf{p}) + \mathcal{W}(\mathbf{d}_{B \to A}; \mathbf{d}_{A \to B})(\mathbf{p}) \Big\|_1. \tag{5.14}$$

Cyclic consistency is especially valuable in the semi-supervised or unsupervised setting, where ground-truth displacements are absent Meister et al. [2018], Jonschkowski et al. [2020]. It can also be used to generate per-pixel reliability maps: pixels where the forward–backward residual exceeds a threshold are likely occluded or unreliable.

5.4 Mask and Confidence Losses

The predicted confidence mask $\hat{c}$ indicates the network's belief that the displacement estimate at each pixel is trustworthy. Supervision of this output can be explicit or self-supervised.

5.4.1 Supervised Mask Loss

When a ground-truth mask $c^* \in \{0,1\}^{H \times W}$ is available (e.g., marking regions of sensor saturation, occlusion, or failed annotation), a standard binary cross-entropy (BCE) loss suffices:

$$\mathcal{L}_{\text{mask}} = -\frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \Big[c^* \log \hat{c} + (1 - c^*) \log(1 - \hat{c}) \Big]. \tag{5.15}$$

5.4.2 Self-Supervised Confidence via Learned Attenuation

In many cases no ground-truth confidence map exists. A principled self-supervised alternative, introduced by Kendall and Gal [2017] for heteroscedastic aleatoric uncertainty, modulates the reconstruction loss by the predicted confidence and adds a regularising log-barrier:

$$\mathcal{L}_{\mathrm{conf}} = \frac{1}{|\Omega|} \sum_{\mathbf{p}\in\Omega} \left[\frac{\mathcal{L}_{\mathrm{recon}}(\mathbf{p})}{\hat{c}(\mathbf{p})} + \log \hat{c}(\mathbf{p}) \right], \tag{5.16}$$

where $\mathcal{L}_{\mathrm{recon}}$ is any pixel-wise reconstruction loss (Section 5.2). The first term incentivises the network to assign *low* confidence (small $\hat{c}$) where the reconstruction error is large; the second term prevents the degenerate solution of setting $\hat{c} \to 0$ everywhere, because $\log \hat{c} \to -\infty$. The equilibrium teaches the network to "explain away" genuinely difficult regions (sensor noise, modality-specific textures) rather than distorting the displacement field to accommodate them.

Equivalently, if the network outputs a log-variance $s = \log \sigma^2$ instead of a direct confidence, Eq. (5.16) becomes the negative log-likelihood of a Gaussian with learned variance, providing a Bayesian motivation for the attenuation trick.

5.5 Total Loss

The composite training objective is a weighted sum of the individual terms:

$$\mathcal{L} = \lambda_{\mathrm{sup}} \mathcal{L}_{\mathrm{ms}} + \lambda_{\mathrm{photo}} \mathcal{L}_{\mathrm{photo}} + \lambda_{\mathrm{smooth}} \mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{fold}} \mathcal{L}_{\mathrm{fold}} + \lambda_{\mathrm{cyc}} \mathcal{L}_{\mathrm{cyc}} + \lambda_{\mathrm{conf}} \mathcal{L}_{\mathrm{conf}}, \tag{5.17}$$

where $\mathcal{L}_{\mathrm{photo}}$ is a combination of SSIM, NCC, MIND, perceptual, and edge losses whose relative weights are tuned by ablation (Section 5.7). Loss weights $\{\lambda_{(\cdot)}\}$ are hyperparameters; we initialise them such that each term contributes roughly equally to the total gradient magnitude at the start of training, then hold them fixed. An alternative is to learn the weights using homoscedastic task uncertainty Kendall et al. [2018], though we find manual tuning sufficient for our setting.

5.6 Training Strategy

5.6.1 Optimiser and Learning Rate

We use **AdamW** Loshchilov and Hutter [2019] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}). The learning rate follows a **OneCycleLR** schedule Smith and Topin [2019]: a linear warm-up from $\eta_{\min} = 10^{-6}$ to $\eta_{\max} = 4 \times 10^{-4}$ over the first 10% of iterations, followed by cosine annealing back to $\eta_{\min}$. OneCycleLR consistently outperforms both step-decay and plain cosine schedules in our experiments, echoing findings in the optical flow literature Teed and Deng [2020].

5.6.2 Batch Size and Memory Considerations

Correlation-volume–based architectures (e.g., RAFT) construct 4D cost volumes of size $H \times W \times H \times W$ (or sampled variants thereof). At training resolution 480×640, even the all-pairs correlation in RAFT's efficient implementation consumes several gigabytes per sample. We use a batch size of 6 on 40 GB A100 GPUs, employing mixed-precision (`torch.cuda.amp`) and gradient checkpointing to reduce peak memory. If a larger effective batch is needed (e.g., for stable batch-normalisation statistics), gradient accumulation over two or more micro-batches is preferred over reducing resolution.

5.6.3 Pretraining on Synthetic Optical Flow Data

Dense ground-truth displacement fields are expensive to obtain for real IR–RGB pairs. Pretraining on large-scale synthetic optical flow datasets—**FlyingChairs** Dosovitskiy et al. [2015], **FlyingThings3D** Mayer et al. [2016a], and **Sintel** Butler et al. [2012]—provides the encoder with a strong prior for correspondence estimation. The pretraining schedule follows the "C+T" protocol of RAFT: train on FlyingChairs for 120k iterations, then FlyingThings3D for 120k iterations.

After pretraining, we **fine-tune** on the IR–RGB dataset. During fine-tuning, the photometric and modality-specific losses (Section 5.2) are activated, and the learning rate is reduced by a factor of 5. We observe that pretraining reduces final EPE by 20–30% relative to training from scratch, confirming that generic correspondence priors transfer effectively to the cross-modal setting.

5.6.4 Curriculum Training

Real-world IR–RGB misalignments range from sub-pixel parallax to tens of pixels of offset caused by mechanical vibration or thermal expansion. We adopt a *curriculum* strategy Bengio et al. [2009]: during the first third of fine-tuning, the training set is restricted to pairs with ground-truth EPE < 5 px; during the second third, the threshold rises to 15 px; for the final third, all pairs are included. Alternatively, when ground-truth magnitude is unknown, we generate synthetic warp augmentations of increasing magnitude. Curriculum training reduces early instability and improves final accuracy on large-displacement cases by approximately 8% EPE.

5.6.5 Data Augmentation

We apply the following augmentations, drawn from the RAFT training protocol Teed and Deng [2020]:

- **Spatial:** random crops (400×720), horizontal and vertical flips, small affine perturbations (rotation $\pm 5°$, scale 0.9–1.1).
- **Photometric (applied independently to each modality):** brightness/contrast jitter, Gaussian noise (σ up to 0.04), and random gamma correction.
- **Synthetic displacement augmentation:** a thin-plate-spline warp with random control point displacements sampled from $\mathcal{N}(0, \sigma^2_{\text{tps}})$, applied to one image while the displacement field is updated accordingly. This augmentation effectively increases dataset diversity and has been shown beneficial for flow generalisation Aleotti et al. [2021].

5.6.6 Validation and Checkpoint Selection

Sequence-level splits. To avoid *temporal leakage*—where adjacent, near-duplicate frames appear in both training and validation—we partition at the **sequence level**, not the frame level. Approximately 15% of video sequences (covering diverse scenes, times of day, and weather conditions) are reserved for validation, and an additional 15% for testing.

Early stopping. We monitor validation EPE after every epoch. If it fails to improve for 15 consecutive epochs, training is halted. The checkpoint with the lowest validation EPE is selected for final evaluation, providing a principled guard against overfitting.

5.7 Experimental Design and Evaluation

5.7.1 Ablation Studies

A systematic set of ablations isolates the contribution of each design choice:

1. **Loss function comparison.** Train with (a) L2 only, (b) L1 only, (c) Charbonnier, (d) Charbonnier + photometric composite. Report EPE on the validation set.
2. **Photometric loss variants.** Compare SSIM, NCC, MIND, perceptual loss, and edge loss in isolation and in combination.
3. **With/without confidence mask.** Evaluate whether the learned attenuation mechanism (Eq. 5.16) improves displacement accuracy in reliable regions.
4. **Multi-scale supervision.** Compare single-scale versus the γ-weighted multi-scale loss of Eq. (5.5).
5. **Regularisation terms.** Measure the effect of smoothness, folding, and cyclic-consistency penalties individually and jointly.
6. **Augmentation strategies.** Ablate spatial augmentations, photometric augmentations, and synthetic displacement augmentation in turn.
7. **Pretraining.** Compare from-scratch training against the C+T pretraining protocol of Section 5.6.3.

5.7.2 Baseline Comparisons

We benchmark against four representative baselines spanning classical and learned paradigms:

Global homography (OpenCV). Detect ORB keypoints, match with brute-force Hamming distance, and estimate a homography via RANSAC (`cv2.findHomography`). This is a zero-parameter baseline that captures only global affine-plus-projective misalignment.

SuperPoint + SuperGlue. A learned feature-matching pipeline DeTone et al. [2018], Sarlin et al. [2020] followed by thin-plate-spline interpolation to produce a dense field. This represents the state of the art in sparse cross-modal matching.

RAFT (pretrained). Off-the-shelf RAFT Teed and Deng [2020] trained on FlyingThings3D, evaluated zero-shot on IR–RGB pairs. This isolates the benefit of domain-specific fine-tuning.

VoxelMorph. A U-Net–based registration network Balakrishnan et al. [2019] originally designed for medical image registration, retrained on our data. This represents an alternative dense registration architecture.

5.7.3 Quantitative Metrics

- **End-point error (EPE):** the primary accuracy metric (Eq. 5.6), reported in pixels.
- **SSIM post-warp:** SSIM between the warped IR image and the RGB reference (on grayscale), measuring the structural quality of alignment.
- **PSNR on edge maps:** PSNR computed between Sobel edge maps of the warped IR and RGB images, providing a modality-invariant signal quality indicator.
- **Mask IoU:** Intersection-over-union between the predicted confidence mask (thresholded at 0.5) and a ground-truth reliability mask, where available.
- **Percentage of correct keypoints (PCK):** fraction of pixels with EPE below a threshold τ (we report $\tau \in \{1, 3, 5\}$ px).

5.7.4 Qualitative Evaluation

Numerical metrics alone cannot capture all alignment artefacts. We provide four forms of qualitative evaluation:

1. **Colour overlay:** the warped IR image is shown in one colour channel (e.g., red) and the RGB image in the complementary channels (green/blue). Misalignment appears as colour fringing at object boundaries.
2. **Checkerboard blending:** alternating tiles from the warped IR and the RGB image, making local misalignment immediately perceptible at tile boundaries.
3. **Displacement field visualisation:** the predicted displacement field is colour-coded using the Middlebury flow colour wheel Baker et al. [2011], with magnitude mapped to saturation and direction to hue.
4. **Confidence mask overlay:** the predicted confidence mask is displayed as a semi-transparent heat map over the aligned pair, highlighting regions the network deems unreliable.

These visualisations are presented for cherry-picked successes, typical cases, and failure modes to give the reader a comprehensive understanding of model behaviour.

5.8 Summary

The loss landscape for cross-modal dense displacement prediction is necessarily composite. Supervised losses provide the strongest direct gradient when ground-truth fields are available, while robust variants (Charbonnier, Huber) guard against annotation noise. Photometric losses bridge the modality gap through structure-aware comparisons (SSIM, NCC, MIND, perceptual features, edge maps) and enable semi-supervised learning. Regularisation losses embed physical priors—smoothness, diffeomorphism, cyclic consistency—that prevent degenerate solutions. The learned attenuation mechanism for confidence prediction elegantly couples uncertainty estimation with the reconstruction objective. These components, combined with a carefully staged training protocol (synthetic pretraining, curriculum learning, sequence-level validation) and thorough ablation-driven experimental design, form the methodological foundation for the alignment system evaluated in subsequent chapters.

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